

Categories, Allegories

Freyd & Scedrov, chapter 1 from §1.4 —
definitions, diagrams, theorems

Conventions

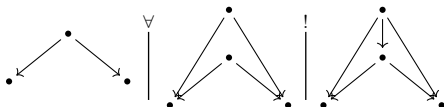
Composition is written by juxtaposition in diagram order: xy is “first x , then y ”, so $\square x$ is the source of x and $x\square$ its target. Bold $\mathbf{A}$ is a category, plain $A, B, \dots$ its objects; $|\mathbf{A}|$ the objects, (A, B) the morphisms from A to B , $\mathbf{A}^\circ$ the opposite, $\mathbf{A}/B$ the slice over B with forgetful $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$, $\mathcal{S}$ the category of sets, $\mathcal{S}^{\mathbf{A}}$ the functor category. 1 is a terminator, 0 an initial object (coterminator), $p_A : A \rightarrow 1$. Products come with $\ell : A \times B \rightarrow A$, $\varkappa : A \times B \rightarrow B$, pairing $\langle f, g \rangle$ and $x \times y$.

Diagrams. Freyd’s own figures, in his diagrammatic language. Every diagram states equations unless punctured. A diagrammatic sentence is a row of diagrams separated by bars carrying $\forall, \exists, \exists!$ or $!$ (“there is at most one extension to the next diagram”); its vertices are bare dots, since the language names nothing.

$+$	puncture mark — removes exactly one equation: that part of the diagram need not commute
$\times$	this wedge is a product diagram
$\sqsubset$	this vertex is a pullback, or a monic pair
$\rhd \rightarrow$	monic
$A \longrightarrow \rhd B$	cover (§1.512) — the hollow triangle head
$\dashrightarrow$	the morphism whose existence and uniqueness is being asserted
$\hookrightarrow$	a member of a distinguished class (dense monics, inclusions)

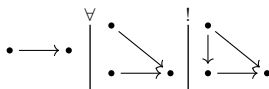
1.4 Cartesian categories

- 1.41 **MONIC PAIR** — $!$ reads “there is at most one extension to”:



in symbols: $\Box x = \Box y \wedge \forall_{u,v}[(ux = vx) \wedge (uy = vy)] \implies u = v$.

a single **MONIC** x , written $A \rightarrowtail B$ — variously called monomorphism, mono, injection; here always monic:



- 1.412 monic family: $\forall_{x \in \mathcal{F}} ux = vx \implies u = v$. A **TABLE** $\langle T; x_1, \dots, x_n \rangle$ is the **TOP** T with a monic finite sequence out of it; the targets are the **FEET**, the x_i the **COLUMNS**.

an isomorphism class of tables is a **RELATION**, $\mathcal{Rel}(A_1, \dots, A_n)$ the family of them on given feet.
 $n = 1$: a **SUBOBJECT** of A , family $\mathcal{Sub}(A)$. $n = 0$: a **VALUE**, family $\mathcal{Val}$.

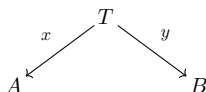
SUB-TERMINATOR — the top of a columnless table:



- 1.413 **CONTAINED**, $\langle T; x_i \rangle \subset \langle T'; x'_i \rangle$: some $z : T \rightarrow T'$ with $zx'_i = x_i$ — necessarily unique and monic. Containment pre-orders tables and partially orders relations, mutual containment being isomorphism; $\mathcal{Rel}$, $\mathcal{Sub}$, $\mathcal{Val}$ are posets.

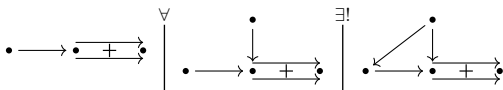
- 1.414 $A \rightarrowtail B$ is monic in $\mathbf{A}$ iff it is a sub-terminator of $\mathbf{A}/B$. Hence $\mathcal{Sub}_{\mathbf{A}}(B) \simeq \mathcal{Val}_{\mathbf{A}/B}$.

1.426



is a table iff $\langle x, y \rangle : T \rightarrow A \times B$ is monic, so
 $\mathcal{R}el(A, B) \simeq \mathcal{S}ub(A \times B)$, generally
 $\mathcal{R}el(A_1, \dots, A_n) \simeq \mathcal{S}ub(A_1 \times \dots \times A_n)$, and $\mathcal{V}al = \mathcal{S}ub(1)$.

- 1.428 **EQUALIZER** diagram — the puncture mark removes just the one equation, so the square carrying it need not commute:

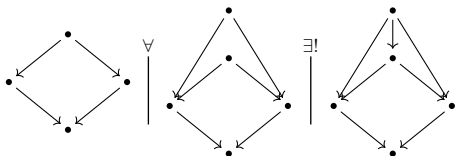


equalizers are monic, so they represent subobjects, and are essentially unique. In $\mathcal{S}$: $\{x \mid fx = gx\}$.

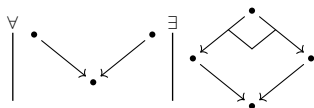
- 1.429 if $e, 1_A$ has an equalizer for every idempotent e , then *all idempotents split*.

- 1.43 **CARTESIAN CATEGORY**: finite products and equalizers (elsewhere: finitely complete, left-exact).

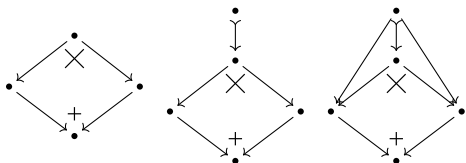
- 1.431 **PULLBACK** diagram — its upper half is a monic pair, and the relation that pair represents is fixed by the lower half:



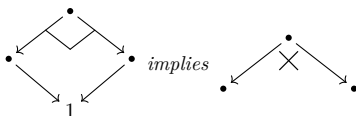
a category has pullbacks if — the corner mark $\lrcorner$ standing for the whole sentence above:



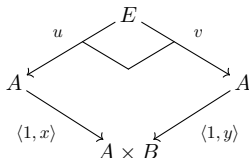
- 1.432 *Binary products + equalizers $\Rightarrow$ pullbacks* — construct in sequence; the outer diagram on the right is the pullback:



1.433 *Pullbacks + a terminator $\Rightarrow$ binary products:*



1.434 *Binary products + pullbacks $\Rightarrow$ equalizers — given $x, y : A \rightarrow B$, then $u = v$ and u equalizes x, y :*

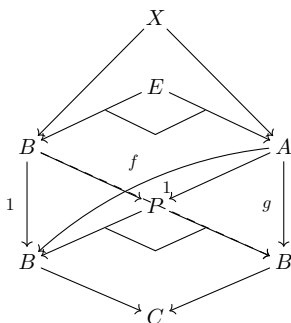


1.435–6 combining the last two: *pullbacks and a terminator imply cartesian*. The three lemmas are exhaustive — all sixteen combinations of terminator / binary products / equalizers / pullbacks not forced by them are realized.

1.437 a functor preserving finite products and equalizers is a **REPRESENTATION OF CARTESIAN CATEGORIES** such preserve pullbacks. Preserving pullbacks + the terminator suffices; so does preserving finite products + pullbacks of monics.

1.438 *A functor reflecting equalizers reflects isomorphisms* ($A \rightarrow B$ is iso iff it equalizes $1_B, 1_B$). From a category with equalizers, an isomorphism-reflecting equalizer-preserving functor is an embedding, hence faithful. A faithful functor preserving terminators / products / equalizers / pullbacks reflects them too.

1.439 *With pullbacks, if $A \xrightarrow{f,g} B \rightarrow C$ then f, g have an equalizer — $P = B \times_C B$ replaces $B \times B$, and $E \rightarrow A$ is the equalizer:*



A functor from a cartesian category preserving pullbacks preserves equalizers.

- 1.44 $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ does not preserve terminators — $\mathbf{A}/B$ has the distinguished terminator 1_B , carried to B — and is universal in that: for $T : \mathbf{C} \rightarrow \mathbf{A}$ with $T(1) = B$ there is a unique $T' : \mathbf{C} \rightarrow \mathbf{A}/B$ with $T'(1) = 1_B$ and $T = T'\Sigma$.

specializing to $T = B \times -$ gives the **DIAGONAL FUNCTOR** $\Delta(A) = (B \times A \xrightarrow{\simeq} B)$ and the factorization $\mathbf{A} \xrightarrow{\Delta} \mathbf{A}/B \xrightarrow{\Sigma} \mathbf{A}$.

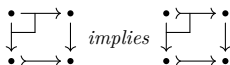
- 1.441 if $\mathbf{A}$ has pullbacks then $\mathbf{A}/B$ is cartesian and Σ preserves pullbacks and equalizers; Σ is always faithful.
- 1.442 the Cayley representation preserves and reflects pullbacks and equalizers; factoring it as $\mathbf{A} \xrightarrow{C'} \mathcal{S}^{|A|} \xrightarrow{\Sigma} \mathcal{S}$ gives a faithful representation. *Every small cartesian category is faithfully represented in a power of $\mathcal{S}$.*

the A -th coordinate of C' is the **REPRESENTABLE FUNCTOR** $(A, -) : B \mapsto (A, B)$, $f \mapsto (A, f)$, i.e. $g \mapsto gf$. *Representable functors from a cartesian category are representations of cartesian categories, and are collectively faithful.*

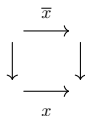
- 1.443 a universal sentence in the pullback / equalizer predicates with a counterexample anywhere has one in $\mathcal{S}$: cut down to a small subcategory, then apply Cayley.

- 1.444 **HORN SENTENCE**: a universally quantified $(P_1 \wedge \dots \wedge P_n) \implies Q$ with all P_i, Q primitive; for cartesian categories the primitives are the categorical ones plus “is a terminator / product / equalizer”. *Any Horn sentence in the theory of cartesian categories true for $\mathcal{S}$ is true for every cartesian category* — apply a collectively faithful $(A, -)$ to the counterexample.

- 1.45 pullbacks transfer monics:

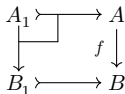


that is, if



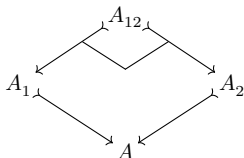
is a pullback and x is monic, then so is $\bar{x}$.

- 1.451 $A_1 \rhd A$ is an **INVERSE IMAGE** of $B_1 \rhd B$ along f :



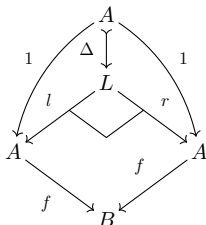
inverse images preserve containment, hence are well defined on subobjects: an order-preserving $f^\# : \mathcal{S}ub(B) \rightarrow \mathcal{S}ub(A)$, making $\mathcal{S}ub(-)$ a contravariant poset-valued functor.

- 1.452 pulling back two monics gives their greatest lower bound in $\mathcal{S}ub(A)$:



so $\mathcal{S}ub(A)$ is a **SEMI-LATTICE** — a poset with finite intersections, the empty one 1_A — and $\mathcal{S}ub(-)$ lands in semi-lattices, inverse images preserving intersections.

- 1.453 **LEMMA** $\mathbf{A}$ cartesian, T preserving pullbacks: T is faithful iff T preserves properness of subobjects — if $A' \succrightarrow A$ is not iso, neither is $TA' \succrightarrow TA$.
- 1.454 the kernel-pair of f measures how far f is from monic: the **LEVEL** of f (elsewhere: kernel-pair, congruence). f is monic iff Δ is iso; this monic $\Delta : A \rightarrow L$ is the **DIAGONAL** morphism, representing the diagonal subobject.



- 1.46 the cartesian structure of the prime examples.
- 1.462 for small $\mathbf{A}$, $\mathcal{S}^{\mathbf{A}}$ is cartesian and the **EVALUATION FUNCTORS** $E_{\mathbf{A}} : \mathcal{S}^{\mathbf{A}} \rightarrow \mathcal{S}$, $E_{\mathbf{A}}(F) = F(\mathbf{A})$, are representations of cartesian categories, collectively faithful. Assembled into the forgetful $\mathcal{S}^{\mathbf{A}} \rightarrow \mathcal{S}^{|\mathbf{A}|}$ they give a faithful representation into a power of $\mathcal{S}$. Hence $F_1 \rightarrow F_2$ is monic iff every $F_1 \mathbf{A} \rightarrow F_2 \mathbf{A}$ is.
- 1.463 $H^{\mathbf{A}}$ denotes $(\mathbf{A}, -)$ viewed in $\mathcal{S}^{\mathbf{A}}$. As a right $\mathbf{A}$ -set it is the sub $\mathbf{A}$ -set of $\mathbf{A}$ generated by $1_{\mathbf{A}}$: for any right $\mathbf{A}$ -set X and $x \in X$ with $x\Box = \mathbf{A}$ there is a unique $H^{\mathbf{A}} \rightarrow X$ sending $1_{\mathbf{A}} \mapsto x$, $y \mapsto xy$. So $(H^{\mathbf{A}}, X) \simeq \{x \in X \mid x\Box = \mathbf{A}\}$, and for the functor F associated to X , $(H^{\mathbf{A}}, F) \simeq F(\mathbf{A})$, naturally — $(H^{\mathbf{A}}, -)$ and $E_{\mathbf{A}}$ are conjugate.
- 1.464 $(H^{\mathbf{A}}, H^{\mathbf{B}}) \simeq (\mathbf{B}, \mathbf{A})$, so each $x : \mathbf{B} \rightarrow \mathbf{A}$ gives $H^x : H^{\mathbf{A}} \rightarrow H^{\mathbf{B}}$ and a contravariant full embedding $H : \mathbf{A} \rightarrow \mathcal{S}^{\mathbf{A}}$, the **YONEDA REPRESENTATION**.
 $\text{Hom} : \mathbf{A}^{\circ} \times \mathbf{A} \rightarrow \mathcal{S}$ sends $\mathbf{A}, \mathbf{B}$ to $(\mathbf{A}, \mathbf{B})$; the composite $\mathbf{A} \times \mathcal{S}^{\mathbf{A}} \xrightarrow{H^{\circ} \times 1} (\mathcal{S}^{\mathbf{A}})^{\circ} \times \mathcal{S}^{\mathbf{A}} \xrightarrow{\text{Hom}} \mathcal{S}$ is naturally equivalent to $E : \mathbf{A}, F \mapsto F(\mathbf{A})$.
- 1.465 $H_{\mathbf{B}}$ denotes the contravariant $(-, \mathbf{B})$ viewed in $\mathcal{S}^{\mathbf{A}^{\circ}}$; again $(H_{\mathbf{B}}, F) \simeq F(\mathbf{B})$, giving a covariant full embedding $\mathbf{A} \rightarrow \mathcal{S}^{\mathbf{A}^{\circ}}$, also called the yoneda

representation. It preserves and reflects the cartesian predicates.

- 1.47 $\mathbf{A}$ is *special* if every universal (not merely Horn) sentence in the cartesian predicates true for $\mathcal{S}$ holds in $\mathbf{A}$.
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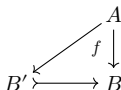
- 1.471–4 *A special cartesian category has at most two values.*
 If every finite sequence of morphisms is preserved-and-reflected by some $T : \mathbf{A} \rightarrow \mathcal{S}$, then $\mathbf{A}$ is special.
 One-valued $\mathbf{A}$: special iff every $B \times -$ is faithful.
 Two-valued $\mathbf{A}$, with 0 the proper subobject of 1:
 special iff $B \times -$ is faithful for every $B \neq 0$.
-

- 1.48 a class $\mathcal{D}$ of monics in a cartesian $\mathbf{A}$ is **DENSE** if it holds the isomorphisms and is closed under composition and pullback. The **RATIONAL CATEGORY** $\mathbf{A}[\mathcal{D}^{-1}]$, with $T_{\mathcal{D}}$, inverts every dense monic and is universal among functors that do.
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- 1.481 $\mathbf{A}[\mathcal{D}^{-1}]$ is cartesian and $T_{\mathcal{D}}$ is a representation of cartesian categories.
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1.5 Regular categories

- 1.51 a subobject $B' \rightarrowtail B$ **ALLOWS** f iff f factors through it:



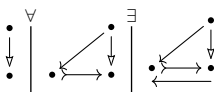
with pullbacks: B' allows f iff $f^\#(B')$ is entire, all of A . A faithful pullback-preserving functor reflects allowance.

the **IMAGE** of f is the smallest subobject allowing f ; a category **HAS IMAGES** if every morphism has one. Then $f : \mathcal{S}ub(A) \rightarrow \mathcal{S}ub(B)$ sends $A' \rightarrowtail A$ to the image of $A' \rightarrowtail A \rightarrow B$.

with pullbacks too, the first adjoint pair of poset functions: $f(A') \subset B'$ iff $A' \subset f^\#(B')$. f is the **LEFT-ADJOINT** of $f^\#$, $f^\#$ the **RIGHT-ADJOINT** of f .

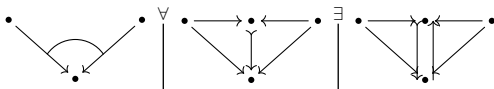
- 1.511 If $\mathcal{A}$ has images and T is faithful and preserves images, then T reflects images.

- 1.512 $A \rightarrow B$ is a **COVER**, written $A \twoheadrightarrow B$, if its image is entire — every monic it factors through is invertible:



covers are closed under composition and left cancellation ($A \rightarrow B \rightarrow C$ a cover forces $B \rightarrow C$ to be one); a monic cover is an isomorphism.

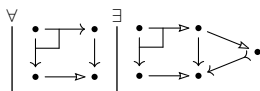
- 1.513 $\{A_i \rightarrow B\}$ **COVERS** if no proper subobject of B allows all of them — the arc across the two arrows says the pair covers:



- 1.514 $\{A_i \rightarrow B\}$ is **EPIC** if its members collectively cancel — a monic family in the opposite category. With

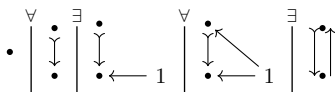
equalizers, cover implies epic; the converse is a special property.

- 1.52 a **REGULAR CATEGORY** is a cartesian category with images in which pullbacks transfer covers — one sentence for both clauses, the right-hand vertex being the image of the right leg:

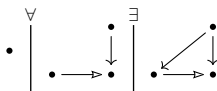


a **PRE-REGULAR CATEGORY** is cartesian, with or without images, and transfers covers by pullback — the strength at which the statements below hold. A **REPRESENTATION OF PRE-REGULAR CATEGORIES** preserves finite products, equalizers and covers, hence whatever images exist.

- 1.521 for small $\mathbf{A}$, $\mathcal{S}^{\mathbf{A}}$ is regular and the evaluation functors are collectively faithful representations of regular categories.
- 1.522 the **SUPPORT** $\mathcal{S}pt(A)$ is the image of $A \rightarrow 1$; A is **WELL-SUPPORTED** if $\mathcal{S}pt(A) = 1$ — pre-regularly, if $A \rightarrow 1$ covers.
- 1.523 A is **WELL-POINTED** if all of $1 \rightarrow A$ covers A : no proper subobject of A allows every point.



- 1.524 P is **PROJECTIVE** if $(P, -)$ preserves covers — every $P \rightarrow B$ lifts through every cover:



Pre-regularly it suffices that every cover targeted at P have a left inverse.

- 1.525 a pre-regular category is **CAPITAL** if every well-supported object is well-pointed. In a capital pre-regular category the terminator is projective.

- 1.526 so for capital pre-regular $\mathbf{A}$ the functor Γ represented by 1 is a representation of pre-regular categories.
- 1.53 **THE SLICE LEMMA** factor $(B \times -)$ as $\mathbf{A} \xrightarrow{\Delta} \mathbf{A}/B \xrightarrow{\Sigma} \mathbf{A}$, Δ the diagonal functor [1.44], $\Delta(1) = \left(B \xrightarrow{1} B \right)$. If $\mathbf{A}$ is (pre-)regular so is $\mathbf{A}/B$; Δ is a representation of pre-regular categories; Δ is faithful iff B is well-supported.
- 1.531 $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ preserves and reflects covers and pullbacks and carries factorizations back, so images in $\mathbf{A}$ give images in $\mathbf{A}/B$.
- 1.535 *Diagonal* also names the morphism $A \rightarrow \Pi_I A$ with identity coordinates; Δ is not that functor but is conjugate to it.
- 1.54 **THE CAPITALIZATION LEMMA** let $\mathcal{C}$ have small pre-regular categories as objects and faithful representations as morphisms.
- 1.541 $\mathcal{C}$ meets the *equivalence condition* if $\mathbf{A} \in \mathcal{C}$ and an equivalence $\mathbf{A} \rightarrow \mathbf{B}$ force $\mathbf{B}, (\mathbf{A} \rightarrow \mathbf{B}) \in \mathcal{C}$; the *slice condition* if $\mathbf{A} \in \mathcal{C}$ and B well-supported force $\mathbf{A}/B, \Delta \in \mathcal{C}$; the *union condition* if a **DIRECTED UNION** $\mathbf{A} = \cup \mathcal{F}$ of $\mathcal{C}$ -subcategories with $\mathcal{C}$ -inclusions has $\mathbf{A}, (\mathbf{A}' \subset \mathbf{A}) \in \mathcal{C}$.
- 1.542 all three hold for the small pre-regular and the small regular categories, the slice condition being [1.53].
- 1.543 Under the three conditions every $\mathbf{A} \in \mathcal{C}$ admits a capital $\underline{\underline{\mathbf{A}}} \in \mathcal{C}$ with $\mathbf{A} \rightarrow \underline{\underline{\mathbf{A}}}$ faithful in $\mathcal{C}$.
- 1.544 to read $\mathbf{A}$ inside $\mathbf{A}/B$, pass to the **INFLATION** $\mathbf{A}'$ — objects finite sequences of objects, binary product concatenation — giving strict cancellation $B \times A = B \times A' \implies A = A'$, then replace the image of Δ by $\mathbf{A}$.
- 1.545 $\mathbf{A} \subset \mathbf{A}^*$ is a **RELATIVE CAPITALIZATION** if every proper $B' \succrightarrow B$ of $\mathbf{A}$, B well-supported, fails to allow some $x : 1 \rightarrow B$ of $\mathbf{A}^*$. Iterated union gives $\underline{\underline{\mathbf{A}}}$; each slicing step supplies ΔB with its generic point, which $\Delta B'$ cannot allow:

$$\begin{array}{ccc}
 & \langle 1, 1 \rangle & \\
 B & \xrightarrow{\quad} & B \times B \\
 & \searrow \quad \swarrow & \\
 & 1 & B
 \end{array}$$

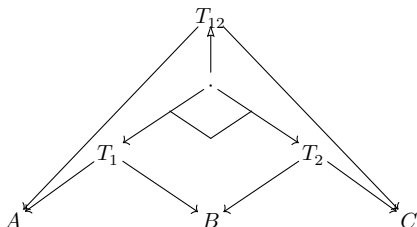
- 1.547 choice-free variant: the rational category [1.48] of pairs $\langle A, F \rangle$, F a finite set of morphisms to distinct well-supported targets — a directed union of slices $\mathbf{A}/\Pi U$.
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- 1.55 **THE HENKIN–LUBKIN THEOREM** *Every small pre-regular category is faithfully represented in a power of the category of sets: $T_B = \mathbf{A} \xrightarrow{\Delta} \mathbf{A}/B \rightarrow \underline{\underline{\mathbf{A}/B}} \xrightarrow{\Gamma} \mathcal{S}$; for regular $\mathbf{A}$ the $\{T_U\}_{U \subset 1}$ already suffice.*
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- 1.551 *Every Horn sentence in the defining predicates of regular categories true for the category of sets is true for every regular category.*
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- 1.552 a pre-regular category is **SPECIAL** if every universally quantified sentence in the relevant predicates true for $\mathcal{S}$ holds in it; the conditions of [1.47] stay necessary. *Special iff $A \rightarrow U \multimap 1$ forces $A \rightarrow U$ or $U \multimap 1$ to be an isomorphism*
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equivalently: one-valued ($|\pi_0 \mathbf{A}| = 1$), or two-valued ($|\pi_0 \mathbf{A}| = 2$) with every object well-supported or isomorphic to 0, the unique proper sub-terminator.

- 1.56 tables T_1 on A, B and T_2 on B, C : a table T_{12} on A, C is a **COMPOSITION** if there exists



cartesian + images builds one exactly so; shrinking either factor shrinks the composite, giving an order-preserving $\mathcal{R}el(A, B) \times \mathcal{R}el(B, C) \rightarrow \mathcal{R}el(A, C)$. *Caution:* not necessarily associative [1.569].

- 1.561 the **RECIPROCAL** R° of R from A to B twists the columns: $R^{\circ\circ} = R$, $(RS)^\circ = S^\circ R^\circ$. Reciprocation reverses composition, preserves the ordering.
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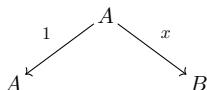
- 1.562 $\mathcal{R}el(A, B) \simeq \mathcal{S}ub(A \times B)$ makes $\mathcal{R}el(A, B)$ a semi-lattice: $(R \cap S)T \subset RT \cap ST$, $(R \cap S)^\circ = S^\circ \cap R^\circ$.

- 1.563 F preserving the cartesian structure and images induces $\mathcal{Rel}(A, B) \rightarrow \mathcal{Rel}(FA, FB)$ preserving composition, reciprocation and intersection; faithful F reflects them.

Hence a Horn sentence on maps and relations — the regular predicates, plus composition, reciprocation and intersection — true of binary relations between sets is true in every regular category.

in particular $R(ST) = (RS)T$, and the **MODULAR IDENTITY** $RS \cap T \subset (R \cap TS^\circ)S$. Every containment is an equation: $R \subset S$ iff $R = R \cap S$.

- 1.564 $\mathcal{Rel}(\mathbf{A})$ is the category of relations; the **GRAPH** of x is tabulated by



giving $\mathbf{A} \rightarrow \mathcal{Rel}(\mathbf{A})$. Treat $\mathbf{A}$ as a subcategory: **MAP** means a morphism of $\mathcal{Rel}(\mathbf{A})$ lying in $\mathbf{A}$, and $\langle T; x, y \rangle$ tabulates one iff x is an isomorphism.

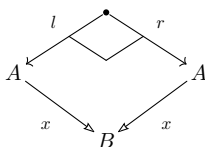
R is **ENTIRE** iff $1 \subset RR^\circ$, **SIMPLE** iff $R^\circ R \subset 1$. For R tabulated by x, y : x covers iff R is entire, x is monic iff R is simple. So R is a map iff entire and simple — hence as soon as every representation $F : \mathbf{A} \rightarrow \mathcal{S}$ makes $F(R)$ one.

- 1.565 a **PUSHOUT** is a pullback in the opposite category; the corner mark moves to the far vertex of the square, C .

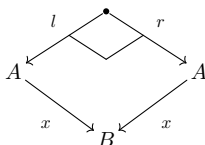
In a regular category a pullback of covers is a pushout — all four edges then cover, the mediating map being the relation $(x^\circ u) \cap (y^\circ v)$, a map because it is one in $\mathcal{S}$:



- 1.566 a **COEQUALIZER** — an equalizer in the opposite category — is always a cover. In a regular category every cover is a coequalizer — of its own level:

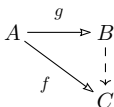


- 1.567 an endo-relation E is an **EQUIVALENCE RELATION** if $1 \subset E$, $E^\circ \subset E$, $EE \subset E$. Levels are equivalence relations:



E is **EFFECTIVE** if it is the level of some map; an **EFFECTIVE REGULAR CATEGORY** is one where every equivalence relation is effective. The metatheorem [1.551] extends to these, $\mathcal{S}$ being effective.

- 1.568 covers out of A are preordered by $f \leq g$ iff f factors through g :



the associated poset is $Quot(A)$, of **QUOTIENT-OBJECTS** — not the dual of $Su\ell(A)$. By [1.566] a faithful order-reversing functor sends it to the poset of equivalence relations on A .

- 1.569 For $\mathbf{A}$ cartesian with images, composition of relations is associative iff $\mathbf{A}$ is regular — x covers iff $x^\circ x = 1$; if z fails to cover, $(yx^\circ)x$ is not even a map.

- 1.56(10) x is a **CONSTANT MORPHISM** if $yx = y'x$ for all $y, y' : C \rightarrow A$. In a regular category its image is a sub-terminator.

- 1.56(11) In a regular category an object is projective iff every entire relation out of it contains a map.

1.57 an object is **CHOICE** if every entire relation targeted at it contains a map. Every object is projective iff every object is choice; an **AC REGULAR CATEGORY** is a regular category with either — equivalently, a cartesian category where every morphism is a left-invertible followed by a monic.

1.571 if $\mathbf{A}$ is cartesian and every $x : A \rightarrow B$ admits $e : A \rightarrow A$ with $e^2 = e$, $ex = x$, then e and x share a level and $\mathbf{A}$ is AC regular — the image of x appears as a subobject of A .

1.572 the category $\mathbf{R}$ of recursive functions on $0, 1, 2, \dots, \omega$ is AC regular but not effective: a left inverse would choose representatives, forcing a recursively enumerable equivalence relation to be recursive.

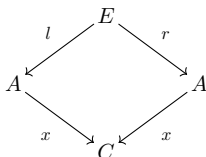
1.58 a **BICARTESIAN CATEGORY** is both cartesian and **COCARTESIAN**, the latter meaning the opposite category is cartesian. Dual to a terminator is a **COTERMINATOR**, written 0 ; dual to a product a **COPRODUCT**, written $A + B$.

in a cartesian category an object 0 whose incoming morphisms are all isomorphisms is a coterminator, then called a **STRICT COTERMINATOR** — agreeing with the earlier 0 [1.474, 1.552].

1.581 between regular cocartesian categories a representation of bicartesian categories is a representation of regular categories: it preserves coequalizers, hence covers [1.566].

1.582 for bicartesian $\mathbf{A}$ regularity is a pair of Horn sentences in the bicartesian predicates — the image of x is $A \twoheadrightarrow C \rightarrowtail B$, with $A \twoheadrightarrow C$ the coequalizer of the level of x .

1.583 effectiveness likewise: for l, r tabulating an equivalence relation and x their coequalizer, E is effective iff



is a pullback.

-
- 1.584 $\mathbf{A}/B$ is cocartesian and Σ a faithful representation of cocartesian categories.
-
- 1.586 for small $\mathbf{A}$, $\mathcal{S}^{\mathbf{A}}$ is cocartesian with collectively faithful evaluation functors, hence effective regular.
-
- 1.587 no elementary characterization of the bicartesian categories obeying every Horn sentence true for $\mathcal{S}$: $1 + \mathbb{N} \xrightarrow{\binom{0}{s}} \mathbb{N}$ iso plus $\mathbb{N} \rightarrow 1$ a coequalizer of $1, s$ pins down $\mathbb{N}, 0, s$, then addition and multiplication; the equalizer of two polynomials is a coterminator iff the equation is unsolvable, so those Horn sentences encode the diophantine problems and are not recursively enumerable.

1.6 Pre-logoi

- 1.6 A **PRE-LOGOS**: a regular category in which $\mathcal{S}ub(A)$ is a lattice, not just a semi-lattice, for each A , and $f^\# : \mathcal{S}ub(B) \rightarrow \mathcal{S}ub(A)$ is a lattice homomorphism for each $f : A \rightarrow B$.

Elementary form: each pair of subobjects has a union, preserved under inverse image.

Equivalently: a cartesian category with images in which pullbacks transfer finite covers. For a cover $\{B_i \rightarrow B\}_{i=1}^n$ and a map $A \rightarrow B$ there exist

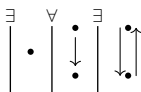
$$\begin{array}{ccc} A_i & \longrightarrow & A \\ \downarrow & & \downarrow \\ B_i & \longrightarrow & B \end{array}$$

$i = 1, \dots, n$, with $\{A_i \rightarrow A\}_{i=1}^n$ a cover. Their existence forces the pullback squares to be such.

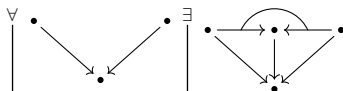
- 1.61 0 = the minimal subobject of 1 ; $p_A^\#(0)$ = the minimal subobject of A . Any morphism to 0 is an isomorphism. Each A receives exactly one $0 \rightarrow A$: 0 is a coterminator.

A pre-logos is degenerate iff it is one-valued, $0 \simeq 1$.

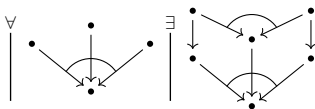
- 1.611 Hence a pre-logos is a cartesian category with images and three sentences. First, 0 exists and every morphism to it is invertible:



second, any two morphisms with a common target have a union — the arc says the pair covers it:



third, inverse image preserves unions — the squares of 1.6 again, now with both pairs covering:



1.612 For monic $f : A \rightarrowtail B$ the inverse image sends $B' \subset B$ to $A \cap B'$.

Every such $f^\#$ preserves binary unions iff $\mathcal{Su}\ell(B)$ is a **DISTRIBUTIVE LATTICE**: $A \cap (B_1 \cup B_2) = (A \cap B_1) \cup (A \cap B_2)$.

1.613 A poset viewed as a category is cartesian iff it is a semi-lattice (intersections are products, equalizers hold by default); a semi-lattice is regular, the only covers being identities. A semi-lattice is a pre-logos iff it is a distributive lattice.

1.614 A **REPRESENTATION OF PRE-LOGOI**: a functor between pre-logoi preserving the cartesian structure, images, and finite unions, the empty union included.

1.615 In a bicartesian category with images the union of $x_1 : A_1 \rightarrowtail A$, $x_2 : A_2 \rightarrowtail A$ is the image of $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} : A_1 + A_2 \rightarrowtail A$.

Building on [1.582], a finite set of Horn sentences in the bicartesian predicates holds for a bicartesian category iff it is a pre-logos.

1.616 $\mathcal{Rel}(A, B) \simeq \mathcal{Su}\ell(A \times B)$ is a distributive lattice, so relations carry $R \cup S$.

$\mathcal{Su}\ell(A \times B)$ distributive	$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$
reciprocation	$(R \cup S)^\circ = S^\circ \cup R^\circ$
$yS = (y \times 1)^\#(S)$, inverse images	$y(S \cup T) = yS \cup yT$
$x^\circ(yS) = (x \times 1)(yS)$, direct images	$R(S \cup T) = RS \cup RT$
reciprocate the line above	$(S \cup T)R' = SR' \cup TR'$

1.62 **PASTING LEMMA** For subobjects A_1, A_2 of A in a pre-logos,

$$\begin{array}{ccc}
 A_1 \cap A_2 & \longrightarrow & A_2 \\
 \downarrow & & \downarrow \\
 A_1 & \longrightarrow & A_1 \cup A_2
 \end{array}$$

is a pushout — the mediating map out of $A_1 \cup A_2$ is unique because A_1, A_2 cover it.

Restated: a pullback whose two morphisms into the lower right corner cover is a pushout.

$$\begin{array}{ccc} \bullet & \xrightarrow{\quad} & \bullet \\ \downarrow & \lrcorner & \downarrow \\ \bullet & \xrightarrow{\quad} & \bullet \end{array} \text{ implies } \begin{array}{ccc} \bullet & \xrightarrow{\quad} & \bullet \\ \downarrow & \lrcorner & \downarrow \\ \bullet & \xrightarrow{\quad} & \bullet \end{array}$$

1.621 If $A_1 \cap A_2 = 0$ and $A_1 \cup A_2 = A$ then A is a coproduct of A_1, A_2 — the special case of the square above.

1.623 A **POSITIVE PRE-LOGOS**: a pre-logos in which every pair A, B admits

$$\begin{array}{ccc} 0 & \xrightarrow{\quad} & B \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{\quad} & C \end{array}$$

Choosing C with $A \cup B = C$, a positive pre-logos has coproducts.

1.624 In a positive pre-logos any $f : A \rightarrow B_1 + B_2$ yields a decomposition, $A_i = f^\#(B_i)$:

$$\begin{array}{ccc} A & \xrightarrow{\sim} & A_1 + A_2 \\ & \searrow f & \downarrow f_1 + f_2 \\ & & B_1 + B_2 \end{array}$$

1.625 For $\mathbf{A}, \mathbf{B}$ positive pre-logoi and $T : \mathbf{A} \rightarrow \mathbf{B}$ a representation of regular categories: T is a representation of pre-logoi iff it preserves disjoint unions.

1.63 $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ yields $\mathcal{S}u\ell_{\mathbf{A}/B}(-) \simeq \mathcal{S}u\ell_{\mathbf{A}}(\Sigma(-))$, respecting inverse images when $\mathbf{A}$ has pullbacks, and unions in $\mathbf{A}$ give unions in $\mathbf{A}/B$.

If $\mathbf{A}$ is a (positive) pre-logos so is $\mathbf{A}/B$, and $\Delta : \mathbf{A} \rightarrow \mathbf{A}/B$ is a representation of pre-logoi.

The capitalization lemma [1.543] applies: any (positive) pre-logos is faithfully representable in a capital (positive) pre-logos.

1.631 $A_1 \subset A$ is a **COMPLEMENTED SUBOBJECT** if some $A_2 \subset A$ has $A_1 \cap A_2 = 0$, $A_1 \cup A_2 = A$ [1.621]; distributivity of $\mathcal{S}u\ell(A)$ makes A_2 unique.

In a positive pre-logos a complemented subobject of a projective object is projective. With [1.525]: *the **COMPLEMENTED SUBTERMINATORS** — complemented subobjects of 1 — in a capital pre-logos are projective.*

- 1.632 $\mathcal{G} \subset |\mathbf{A}|$ is a **GENERATING SET** if $\{(G, -)\}_{G \in \mathcal{G}}$ collectively yields an embedding; $\mathcal{B} \subset |\mathbf{A}|$ is a **BASIS** if $\{(B, -)\}_{B \in \mathcal{B}}$ is collectively faithful. Both elementary.

For cartesian $\mathbf{A}$, $\mathcal{B}$ is a basis iff every proper $A' \twoheadrightarrow A$ admits

$$\begin{array}{ccc} B' & \twoheadrightarrow & B \\ \downarrow & \lrcorner & \downarrow \\ A' & \twoheadrightarrow & A \end{array}$$

with $B \in \mathcal{B}$ and $B' \twoheadrightarrow B$ proper [1.442].

- 1.633 *A positive pre-logos is capital iff the complemented subterminators are projective and form a basis.*

- 1.634 $\mathcal{F} \subset P$ is a **PRE-FILTER** if non-empty and any $x, y \in \mathcal{F}$ have $z \in \mathcal{F}$ with $z \leq x, z \leq y$; a **FILTER** if also an updeal.

For $\mathcal{F}$ a pre-filter in $\mathcal{VAL}_{\mathbf{A}}$: $T_{\mathcal{F}}(A)$ has elements named by $x : U \rightarrow A, U \in \mathcal{F}$, with x and $y : V \rightarrow A$ naming the same element when

$$\begin{array}{ccc} W & \longrightarrow & U \\ \downarrow & & \downarrow x \\ V & \longrightarrow & A \\ & y & \end{array}$$

for some $W \in \mathcal{F}, W \subset U, W \subset V$.

$T_{\mathcal{F}}$ preserves finite products and equalizers; it preserves covers when the elements of $\mathcal{F}$ are projective.

In a pre-logos $T_{\mathcal{F}}$ preserves disjoint unions iff $0 \notin \mathcal{F}$, and $U_1 + U_2 \in \mathcal{F}$ implies $U_1 \in \mathcal{F}$ or $U_2 \in \mathcal{F}$.

- 1.635 **THE REPRESENTATION THEOREM FOR PRE-LOGOI**
Every small positive pre-logos is faithfully representable in a power of the category of sets. ‘Positive’ is removable, every pre-logos being faithfully representable in a positive one [2.217].

In a capital positive pre-logos a representative set $\mathcal{B}$ of complemented subterminators, as a sublattice of $\mathcal{SUB}$,

is distributive with every element complemented — a **BOOLEAN ALGEBRA**.

Every pre-filter $\mathcal{F} \subset \mathcal{B}$ extends, by the axiom of choice, to an **ULTRA-FILTER** $\overline{\mathcal{F}}$: a maximal pre-filter with $0 \notin \overline{\mathcal{F}}$, necessarily a filter.

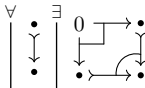
For $\overline{\mathcal{F}}$ an ultra-filter, $T_{\overline{\mathcal{F}}}$ is a representation of pre-logoi. Collectively faithful, because $\mathcal{B}$ is a basis.

1.636 Any Horn sentence in the predicates of pre-logoi true for the category of sets is true for all positive pre-logoi — ‘positive’ again removable [2.217].

1.637 A special pre-logos satisfies every universal sentence in the predicates of pre-logoi satisfied by $\mathcal{S}$. A pre-logos is a special regular category iff two-valued.

A pre-logos is special iff $(A' \times B) \cup (A \times B')$ is a proper subobject of $A \times B$ for every pair of proper $A' \subset A$, $B' \subset B$.

1.64 A **BOOLEAN PRE-LOGOS**: a pre-logos in which $\mathcal{S}ub(A)$ is a boolean algebra for all A . $\mathcal{S}ub(A)$ being distributive, only complements need be required, and they are then unique:



1.641 A representation of pre-logoi between boolean pre-logoi preserves the boolean structure automatically: each $\mathcal{S}ub(A) \rightarrow \mathcal{S}ub(TA)$ preserves complements and is a map of boolean algebras.

1.644 Any power of $\mathcal{S}$ is a boolean capital positive pre-logos; confuse $\mathcal{V}al_{\mathcal{S}^I}$ with the power set of I .

For an ultra-filter $\mathcal{F} \subset \mathcal{V}al_{\mathcal{S}^I}$, $T_{\mathcal{F}}$ is an **ULTRA-PRODUCT FUNCTOR** and the composite an **ULTRA-POWER FUNCTOR**. Both are representations of boolean pre-logoi: $\mathcal{S} \xrightarrow{\Delta} \mathcal{S}^I \xrightarrow{T_{\mathcal{F}}} \mathcal{S}$.

1.645 For $T : \mathbf{A} \rightarrow \mathbf{B}$ a representation of boolean pre-logoi: $\mathcal{K}er(T) = \{U \subset 1 \mid T(U) = 0\}$, the values killed by T ; $prop(A' \subset A)$, the *properness*, is the support of the complement of A' .

$A' \subset A$ is proper iff $\text{prop}(A' \subset A) \neq 0$; $TA' \subset TA$ is proper iff $\text{prop}(A' \subset A) \notin \text{Ker}(T)$; T is faithful iff $\text{Ker}(T) = 0$.

$1 \in \text{Ker}(T)$ iff $\mathbf{B}$ is degenerate, so $\mathbf{A}$ two-valued and $\mathbf{B}$ non-degenerate forces T faithful — every ultra-power functor is faithful. $\text{Ker}(T_{\mathcal{F}})$ is the complement of $\mathcal{F}$ in $\mathcal{Val}_{\mathbf{A}}$.

1.646 Two propositions with the same proof: *every small special cartesian category, and every small special positive pre-logos, is faithfully representable in the category of sets.*

1.647 For proper $A_i \subset B_i$ with complement A'_i ($i = 1, 2$), the complement of $(A_1 \times B_2) \cup (B_1 \times A_2)$ in $B_1 \times B_2$ is $A'_1 \times A'_2$. Hence, by [1.637], *a boolean pre-logos is special iff it is two-valued.*

1.648 If an ultra-power functor preserves coequalizers — one built from $\mathbb{N}, 0, s$ [1.587] suffices — then $\mathcal{F}$ meets every countable partitioning of I : such an ultra-filter is a **COMPLETE MEASURE**.

The **ATOMIC MEASURES** are those of the form $\{j \mid i \leq j\}$; for $\mathcal{F}$ atomic, $T_{\mathcal{F}}$ is conjugate to a projection and the ultra-power functor to the identity functor.

1.65 A **PRE-TOPOS**: an effective positive pre-logos. Every pre-logos is faithfully represented in a pre-topos [2.217].

1.651 **THE AMALGAMATION LEMMA** *Monics with a common source $x : A \rightarrowtail B$, $y : A \rightarrowtail C$ in a pre-topos extend to*

$$\begin{array}{ccc}
 & y & \\
 & \downarrow & \\
 A & \rightarrowtail & C \\
 \uparrow & \lrcorner & \uparrow \\
 x & \downarrow & \bar{x} \\
 & \lrcorner & \\
 B & \rightarrowtail & D \\
 & \bar{y} &
 \end{array}$$

— take D with level $E = l^\circ x^\circ y r \cup 1 \cup r^\circ y^\circ x l$ on $B + C$, for l, r the coproduct injections. The square is a pullback, hence by [1.62] a pushout.

- 1.652 *In a pre-topos covers coincide with epics and monics coincide with cocompacts — every monic is an equalizer, so the image of an epic is entire.*
- 1.653 *For $A \rightarrow B$ and monic $y : A \rightarrowtail C$ in a pre-topos there exist*

$$\begin{array}{ccc}
 A & \xrightarrow{y} & C \\
 \downarrow & \lrcorner & \downarrow \\
 B & \xrightarrow{\quad} & D
 \end{array}$$

— factor $A \rightarrow B$ through its image and stack the square with [1.651].

- 1.654 A pre-topos need not be cocartesian. If it is, the last two sections show its opposite category is regular.
- 1.655 *If $\mathbf{A}, \mathbf{B}$ are pre-topoi and $T : \mathbf{A} \rightarrow \mathbf{B}$ preserves 0, pushouts, finite products and monics, then T is a bicartesian representation.*
- 1.657 In an effective regular category, for R tabulated by

$$\begin{array}{ccc}
 & T & \\
 x \swarrow & & \searrow y \\
 A & & A
 \end{array}$$

and $f : A \rightarrow B$ a coequalizer of x, y : ff° is the minimal equivalence relation containing R .

A pre-topos is cocartesian iff for every endo-relation R there is a minimal equivalence relation containing R . A functor between bicartesian pre-topoi preserves the bicartesian structure iff it is a representation of pre-logoi preserving minimal equivalence relations.

- 1.658 An object A in a pre-logos is **DECIDABLE** if the diagonal $\langle 1, 1 \rangle : A \rightarrowtail A \times A$ has a complement. *Every object of a pre-topos is decidable iff the pre-topos is boolean — pullbacks of complemented subobjects are complemented.*
- 1.66 Choice objects [1.57]: *in any regular category a subobject of a choice object is choice, and a quotient of a choice object is choice — a quotient $x : A \rightarrow B$ [1.568] is also a subobject of A , via a map contained in x° .*

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- 1.661 *In a regular category finite products of choice objects are choice* — an entire relation targeted at a terminator is already a map.
-
- 1.662 *In a pre-topos: (1) binary coproducts of choice objects are choice; (2) $1 + 1$ is choice; (3) the pre-topos is boolean* — are equivalent.
-

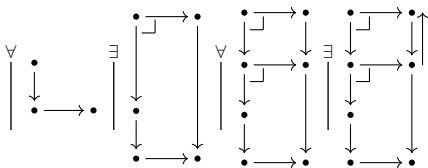
(2) restated by $\mathcal{Rel}(B, 1 + 1) \simeq \mathcal{Sub}(B) \times \mathcal{Sub}(B)$: an entire relation $B \rightarrow 1 + 1$ is a pair of subobjects covering B , a map $B \rightarrow 1 + 1$ a partition of B — so $X \cup Y = B$ in $\mathcal{Sub}(B)$ gives $X' \subset X$, $Y' \subset Y$ with $X' \cup Y' = B$, $X' \cap Y' = 0$.

1.7 Logoi

- 1.7 a **LOGOS** is a regular category in which every $\mathcal{Su}\ell(A)$ is a lattice and every $f^\# : \mathcal{Su}\ell(B) \rightarrow \mathcal{Su}\ell(A)$ has a right adjoint $f^{\#\#} : \mathcal{Su}\ell(A) \rightarrow \mathcal{Su}\ell(B)$: $f^\#(B') \subset A'$ iff $B' \subset f^{\#\#}(A')$.

restated: $f^{\#\#}(A')$ is the maximal subobject $B' \subset B$ with $f^\#(B') \subset A'$.

- 1.71 in $\mathcal{S}$, $f^{\#\#}(A') = \{b \in B \mid \forall_a [f(a) = b \Rightarrow a \in A']\}$. In a boolean pre-logos $f^{\#\#}(A') = \neg(f(\neg A'))$, since $f^\#$ preserves complements.
- 1.711 a logos is a pre-logos: $f^\#$ preserves every union that exists, since $f^\#(\bigcup B_i) \subset A'$ iff every $f^\#(B_i) \subset A'$.
- 1.712 **LOCALLY COMPLETE**: every $\mathcal{Su}\ell(A)$ is a complete lattice. A locally complete regular category whose $f^\#$ preserve all unions is a logos, with $f^{\#\#}(A') = \bigcup \{B_i \mid f^\#(B_i) \subset A'\}$.
- 1.714 the axiom drawn — the corner marks make the squares pullbacks, so each left vertex is the inverse image of the right one; the last panel returns $B'' \subset B'$, so no B'' properly containing B' has $f^\#(B'') \subset A'$.



- 1.72 a **HEYTING ALGEBRA** is a lattice with a binary $x \rightarrow y$ satisfying the double-Horn sentence $z \leq x \rightarrow y$ iff $z \wedge x \leq y$: the largest element meeting x inside y . So $x \rightarrow (_)$ is right adjoint to $x \wedge (_)$.
- as a category a Heyting algebra is a logos: for $x \leq y$ with f the unique morphism, $f^\#(u) = x \wedge u$ and $f^{\#\#}(v) = (x \rightarrow v) \wedge y$.
- 1.721 in a logos $\mathcal{Su}\ell(A)$ is a Heyting algebra: with f the inclusion of A_1 , $(A_1 \rightarrow A_2) = f^{\#\#}(A_1 \cap A_2)$.
- 1.722 a poset viewed as a category is a logos iff it is the poset underlying a (necessarily unique) Heyting algebra.

- 1.723 a **LOCALE** is a complete lattice in which finite intersections distribute over arbitrary unions, $x \wedge \bigvee y_i = \bigvee (x \wedge y_i)$; that is exactly the preservation of arbitrary unions by inverse images, so a locale is a Heyting algebra.

locale morphisms are the union-preserving lattice maps; they need not preserve $\rightarrow$. So ‘complete Heyting algebras’ and ‘locales’ name the same objects, different morphisms.

- 1.724 $x \leftrightarrow y = (x \rightarrow y) \wedge (y \rightarrow x)$, characterized by $z \leq x \leftrightarrow y$ iff $z \wedge x = z \wedge y$: the largest element meeting x and y in the same way. Conversely $x \rightarrow y = x \leftrightarrow (x \wedge y)$.

$\leftrightarrow$ is commutative with unit 1, every element its own inverse ($x \leftrightarrow x = 1$); $x \wedge (x \leftrightarrow y) = x \wedge y$; and it almost distributes with $\rightarrow$: $z \wedge (x \leftrightarrow y) = z \wedge [(z \wedge x) \rightarrow (z \wedge y)]$.

- 1.725 Heyting algebras are equational: to the lattice equations add $x \leftrightarrow 1 = x = 1 \leftrightarrow x$, $x \leftrightarrow x = 1$, and $z \wedge (x \leftrightarrow y) = z \wedge [(z \wedge x) \leftrightarrow (z \wedge y)]$.

1.726	$x \rightarrow (y \rightarrow z) = (x \wedge y) \rightarrow z$; $(x \vee y) \rightarrow z = (x \rightarrow z) \wedge (y \rightarrow z)$; $x \rightarrow (y \wedge z) = (x \rightarrow y) \wedge (x \rightarrow z)$
	$x \rightarrow x = 1$, $1 \rightarrow x = x$, $x \rightarrow 1 = 1$; $x \wedge (x \rightarrow y) = x \wedge y$, $x \leq (x \rightarrow y) \rightarrow y$
	$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$, a case of [1.711]; $x \rightarrow y$ is covariant in y , contravariant in x

- 1.727 the **NEGATION** $\neg x$ is $x \rightarrow 0$, characterized by $z \leq \neg x$ iff $z \wedge x = 0$: the largest element disjoint from x . $\neg$ is contravariant.

special cases of [1.726]: $\neg(x \vee y) = \neg x \wedge \neg y$, $\neg 1 = 0$, $\neg 0 = 1$, $x \leq \neg \neg x$, $\neg x = \neg \neg \neg x$.

double negation $\bar{x} = \neg \neg x$: $x \leq y$ implies $\bar{x} \leq \bar{y}$; $x \leq \bar{x}$, $\bar{\bar{x}} = \bar{x}$, and it preserves intersection, $\overline{x \wedge y} = \bar{x} \wedge \bar{y}$.

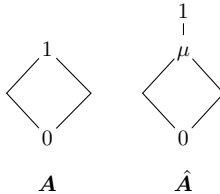
- 1.728 adjoining the **LAW OF EXCLUDED MIDDLE** $x \vee \neg x = 1$ gives every element a complement; Heyting algebras are distributive lattices, so this makes them boolean. Adjoining $x = \neg \neg x$ does the same.

- 1.729 let $f^\# : \mathbf{B} \rightarrow \mathbf{A}$ be order-preserving between Heyting algebras, with a left adjoint f satisfying $b \wedge$

$f(a) = f(f^\#(b) \wedge a)$. Then $f^\#$ preserves $\rightarrow$; conversely, preservation of $\rightarrow$ forces that identity.

hence $Su\mathcal{C}(-)$ on a logos is a contravariant functor with values in Heyting algebras.

- 1.72(10) every Heyting algebra $\mathbf{A}$ is covered by a Heyting algebra, its *scone* $\hat{\mathbf{A}}$, for which $\Gamma = (1, _)$ is a representation of bicartesian categories. $\hat{\mathbf{A}}$ adjoins a new 1 above the old top μ , so $\hat{\mathbf{A}}/\mu \simeq \mathbf{A}$ and 1 is no union of objects properly below it:



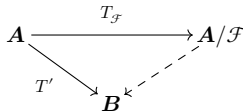
- 1.72(11) for free $\mathbf{A}$ there is a representation of Heyting algebras $\mathbf{A} \rightarrow \hat{\mathbf{A}}$ with $(\mathbf{A} \rightarrow \hat{\mathbf{A}} \rightarrow \hat{\mathbf{A}}/\mu \simeq \mathbf{A}) = 1_{\mathbf{A}}$: $\mathbf{A}$ is a **RETRACT** of its scone.

then Γ is a representation of bicartesian categories on $\mathbf{A}$ too; in particular every finite collection of objects whose join is 1 contains 1.

- 1.73 for a representation of logoi $T : \mathbf{A} \rightarrow \mathbf{B}$, $\mathcal{F}il(T) \subset \mathcal{Val}_{\mathbf{A}}$ collects the values U with $T(U) = 1$; it is a filter.

with $p_A : \mathbf{A} \rightarrow 1$: $A' \subseteq A$ iff $p_A^{\#\#}(A') \subsetneq 1$, and $T(A') \subsetneq T(A)$ iff $p_A^{\#\#}(A') \notin \mathcal{F}il(T)$. T is faithful iff $\mathcal{F}il(T)$ is trivial, $\mathcal{F}il(T) = \{1\}$.

- 1.731 every filter $\mathcal{F} \subset \mathcal{Val}_{\mathbf{A}}$ gives a logos $\mathbf{A}/\mathcal{F}$ and a representation $T_{\mathcal{F}}$ with $\mathcal{F}il(T_{\mathcal{F}}) = \mathcal{F}$, universal among T' with $\mathcal{F} \subset \mathcal{F}il(T')$:



the mediator is unique up to conjugation, and faithful iff $\mathcal{F} = \mathcal{F}il(T')$.

cartesian / regular / pre-logos $\mathbf{A}$ give cartesian / regular / pre-logos $\mathbf{A}/\mathcal{F}$.

1.732 logoi meet the capitalization lemma's union, equivalence and slice conditions: Σ matches $\mathcal{Sub}_{\mathbf{A}/\mathbf{A}}$ with $\mathcal{Sub}_{\mathbf{A}}(\Sigma(_))$ and preserves inverse images, so double-sharps descend to $\mathbf{A}/\mathbf{A}$; Δ preserves them because $A \times (_)$ does.

1.733 A in a pre-logos is **COPRIME** if $(A, _)$ preserves finite unions: any finite collection of subobjects of A whose union is A contains A . A is **CONNECTED** if it has exactly two complemented subobjects.

a **FOCAL LOGOS** is a logos whose terminator is a coprime projective, i.e. one for which $\Gamma = (1, _)$ is a representation of pre-logoi. *a positive pre-logos is focal iff its terminator is a connected projective.*

1.734 $\mathbf{A} \rightarrow \mathbf{F}$ is a **FOCAL REPRESENTATION** if $\mathbf{F}$ is focal, i.e. $\mathbf{A} \rightarrow \mathbf{F} \xrightarrow{\Gamma} \mathcal{S}$ is a representation of pre-logoi.

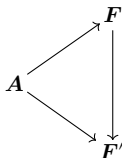
any small (positive) logos has a small, collectively faithful family of focal representations.

a logos may be faithfully represented in a single focal logos iff its terminator is coprime. ('Positive' is removable by [2.217, 2.343].)

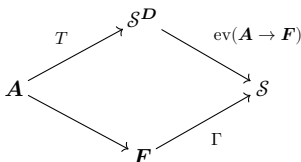
1.735 capitalization raises no cardinality unless the category is finite, and positivization preserves infinite cardinalities. *any countable (positive) logos has a countable, collectively faithful family of focal representations; a countable logos with coprime terminator is faithfully representable in a countable focal logos.*

1.74 **GEOMETRIC REPRESENTATION THEOREM** *every countable (positive) logos may be faithfully represented in a countable power of the logos of sheaves on the real line; with a coprime terminator, in that logos itself.*

1.742 *for any small (positive) logos $\mathbf{A}$ there is a small category $\mathbf{D}$ and a faithful representation $\mathbf{A} \rightarrow \mathcal{S}^{\mathbf{D}}$. Objects of $\mathbf{D}$: focal representations $\mathbf{A} \rightarrow \mathbf{F}$; morphisms: the focal $\mathbf{F} \rightarrow \mathbf{F}'$ under $\mathbf{A}$.*



$$T(A)(A \rightarrow F) = F(1, A):$$



D is enlarged until the morphisms out of each of its objects are collectively faithful.

1.743 $\mathcal{S}^D$ is focal iff D has a coterminator; e.g. when A is focal.

1.744 a functor $D' \rightarrow D$ induces a representation of prelogoi $\mathcal{S}^D \rightarrow \mathcal{S}^{D'}$ by composition, faithful when the functor is onto on objects.

D' **DOMINATES** D if some $F : D' \rightarrow D$ is onto on objects and **LEFT-FULL**: every $g : F(D) \rightarrow B$ in D is $F(f)$ for some $f : D \rightarrow A$ in D' . Such F need not be full.

if D' dominates D then $\mathcal{S}^D$ is faithfully representable in $\mathcal{S}^{D'}$ as a logos.

1.745 *every category (with a coterminator) is dominated by a (rooted) tree: $P(D)$, the finite nonempty paths of composable morphisms ordered by prolongation on the right, each sent to the target of its last map.*

1.746 D^+ , the tree of finite words of morphisms with the greedy functor $D^+ \rightarrow P(D)$, is homogeneous and dominates D . *every countable category with a coterminator is dominated by the binary tree 2^* : finite words on two symbols, ordered by prolongation on the right.*

1.74(10) the **FREYD CURVE** maps $[-1/2, 1/2]$ onto $2^* \cup 2^{\mathbb{N}}$, the finite and infinite binary words with the updeals of the finite words as basic opens.

write $x = \sum_i a_i/3^i$ with $a_i = -1, 0, +1$; delete the zeros to leave $\{a_{n_i}\}_{i=1}^L$, $L \leq \infty$; send x to $\{(-1)^{n_{i-1}}a_{n_i}\}_{i=1}^L$, taking $n_0 = 0$ and reading 2 as $\{-1, +1\}$. Open, continuous and onto, which is what [1.74] needs.

1.75 **STONE REPRESENTATION THEOREM** *for every small (positive) logos $\mathbf{A}$ there is a space X and a faithful representation $\mathbf{A} \rightarrow Sh(X)$; X may be taken compact, totally disconnected and without isolated points, and for countable $\mathbf{A}$ it may be taken to be the Cantor space, the rationals, or the irrationals.*

1.751 an object is an **ATOM** if 0 is its unique proper subobject. A logos is **ATOMICALLY BASED** if its atoms form a basis, which forces it boolean; it is **ATOMLESS** if it has no atoms.

the periodic power $\Pi\mathbf{A}$ (the periodic functions $\mathbb{N} \rightarrow \mathbf{A}$) is a sublogos of $\mathbf{A}^{\mathbb{N}}$ with the diagonal $\mathbf{A} \rightarrow \Pi\mathbf{A}$ faithful; it is atomless for non-degenerate $\mathbf{A}$ and stays positive and capital. So: positive capital atomless logoi.

1.752 $\mathcal{B}$, the boolean algebra of complemented subterminators of $\mathbf{A}$, is a basis, and atomless when $\mathbf{A}$ is.

the **STONE SPACE** $\hat{\mathcal{B}}$ has the ultrafilters on $\mathcal{B}$ as points and the $\hat{U} = \{\mathcal{F} \mid U \in \mathcal{F}\}$, $U \in \mathcal{B}$, as basic opens; it is compact and totally disconnected.

a boolean algebra is isomorphic to the lattice of **CLOPENS** (sets both open and closed) of its Stone space, so its atoms are the isolated points; $\hat{\mathcal{B}}$ here has none.

1.761 $\mathcal{F}$ closed under countable intersection: $\mathbf{A}/\mathcal{F}$ inherits $\mathbf{A}$'s countable unions and coproducts and $\mathbf{A} \rightarrow \mathbf{A}/\mathcal{F}$ preserves them; if closure fails at some size, so does preservation of unions of that size. Closed under arbitrary intersection: $U \in \mathcal{F}$ iff $\bigcap \mathcal{F} \subset U$, and $\mathbf{A}/\mathcal{F} = \mathbf{A}/(\bigcap \mathcal{F})$.

1.77 for a binary endo-relation R in a regular category, the **TRANSITIVE CLOSURE** $R^\dagger$ is the minimal transitive relation containing R , and the **TRANSITIVE-REFLEXIVE CLOSURE** R^* the minimal transitive-reflexive one. In a pre-logos each

exists iff the other does: $R^* = 1 \cup R^\dagger$ and $R^\dagger = RR^*$.

a **TRANSITIVE LOGOS**, likewise a **TRANSITIVE PRE-LOGOS**, is one in which every endo-relation has a transitive closure.

1.771 countable unions in every $\mathcal{S}u\mathcal{B}$, preserved by inverse images (as a logos forces): $R^\dagger = R \cup R^2 \cup \dots \cup R^n \cup \dots$, transitive because composition then distributes with countable union.

1.772 a **Σ -TRANSITIVE LOGOS** is a transitive logos in which $R^\dagger$ is the least upper bound of the finite powers of R ; a **Σ -TRANSITIVE PRE-LOGOS** wants that union stable: $R^\dagger = \bigcup_{n=1}^{\infty} R^n$, preserved under inverse images.

σ -transitivity is an infinitary Horn sentence: $(R \subset T) \wedge (R^2 \subset T) \wedge \dots$ implies $R^\dagger \subset T$; for pre-logoi, $f^\#(R^n) \subset A'$ for all n implies $f^\#(R^\dagger) \subset A'$. Its models are closed under products and substructures, and $\mathcal{S}$ is σ -transitive, so representing transitive pre-logoi in a power of $\mathcal{S}$ requires it.

1.773 *every locally complete regular category is transitive* ($R^\dagger$ is the intersection of all transitive relations containing R), but need not be σ -transitive: $R_1 = R$, $R_{\alpha+1} = R_\alpha \cup (R_\alpha)^2$, $R_\beta = \bigcup_{\alpha < \beta} R_\alpha$ at limits.

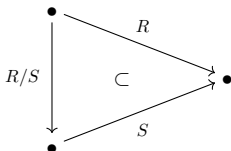
1.775 the **EQUIVALENCE CLOSURE** $R^\equiv$ is the minimal equivalence relation containing R ; in a transitive pre-logos $R^\equiv = (R \cup R^\circ)^*$. An effective regular category has all equivalence closures iff it has coequalizers.

a pre-logos is **E-STANDARD** if every R has an equivalence closure and $R^\equiv$ is always the stable union of the finite powers of $R^\circ \cup 1 \cup R$. σ -transitive implies E-standard, and E-standardness is necessary for a faithful family of set-valued representations preserving equivalence closures.

1.776 *every countable positive σ -transitive pre-logos is faithfully representable in a power of $\mathcal{S}$; every countable positive σ -transitive logos in $\mathcal{S}h(X)$ with X the rationals or the irrationals; every countable E-standard bicartesian pre-topos in a power of $\mathcal{S}$.*

the last characterizes the countable bicartesian categories representable in a power of $\mathcal{S}$. ‘Positive’ is removable [2.217, 2.343]; ‘countable’ is not.

- 1.78 for relations R, S with a common target, R/S is the maximum relation T , if it exists, with $TS \subset R$: $T \subset R/S$ iff $TS \subset R$. The triangle only semi-commutes, as the containment sign says.



- 1.781 $\text{Rel } x(R/S)y \text{ iff } \forall_z (ySz) \Rightarrow (xRz)$.
- 1.782 for a map f with the same target as R : R/f exists and equals Rf° .
- 1.783 $(R/S_1)/S_2 = R/(S_2S_1)$: if the left side exists so does the right, and they agree.
- 1.784 in a logos R/S exists for every pair of relations with a common target. Every S is $l^\circ r$, reducing the question to R/f° ; there $Tf^\circ = (1 \times f)^\#(T)$, so $R/f^\circ = (1 \times f)^{\#\#}(R)$.
- 1.785 conversely, read $A' \succrightarrow A$ followed by the diagonal $\langle 1, 1 \rangle : A \rightarrow A \times A$ as a relation on A : if A'/f° exists then $f^{\#\#}(A') = 1_B \cap (A'/f^\circ)^\circ(A'/f^\circ)$.
- 1.786 for R, S, T with a common target, $(R/S)(S/T) \subset R/T$; and R/R is transitive and reflexive.
- 1.787 $\overline{R}$ is the minimum reflexive relation with $R\overline{R} \subset \overline{R}$. in a logos $\overline{R} = R^*$: if either exists so does the other, and they are equal.

1.8 Adjoint functors, Grothendieck topoi, and exponential categories

- 1.81 **ADJOINT PAIR OF FUNCTORS** $F : \mathbf{A} \rightarrow \mathbf{B}$, $G : \mathbf{B} \rightarrow \mathbf{A}$: some natural equivalence $(FA, B) \simeq (A, GB)$ exists, natural in both variables — as functors $\mathbf{A}^\circ \times \mathbf{B} \rightarrow \mathcal{S}$. F a **LEFT-ADJOINT** of G , G a **RIGHT-ADJOINT** of F .

$F \dashv G$	$(FA, B) \simeq (A, GB)$
in posets	$FA \leq B \text{ iff } A \leq GB$
$\mathbf{A}' \subset \mathbf{A}$ reflective	$(\bar{A}, B) \simeq (A, B), \quad B \in \mathbf{A}'$
$\mathbf{A}' \subset \mathbf{A}$ coreflective	$(B, GA) \simeq (B, A), \quad B \in \mathbf{A}'$
adjoint on the right	$(B, FA) \simeq (A, GB), \quad F, G$ contravariant
adjoint on the left	$(FA, B) \simeq (GB, A), \quad F, G$ contravariant

- 1.811 poset hom-sets have at most one element, so adjointness is $(FA, B) \neq \emptyset$ iff $(A, GB) \neq \emptyset$. Direct image is a left-adjoint of inverse image $f^\#$.

- 1.813 **REFLECTIVE SUBCATEGORY** $\mathbf{A}' \subset \mathbf{A}$: the inclusion has a left-adjoint, whose values are **REFLECTIONS**. Elementary form — every A has $A \rightarrow \bar{A}$, $\bar{A} \in \mathbf{A}'$, and every $A \rightarrow B$, $B \in \mathbf{A}'$, factors by a unique $(\bar{A} \rightarrow B) \in \mathbf{A}'$:

$$\begin{array}{ccc}
 & & \bar{A} \\
 & \nearrow & \vdots \\
 A & & \downarrow \\
 & \searrow & B
 \end{array}$$

one $A \rightarrow \bar{A}$ chosen per A makes reflection a functor: $\bar{x}$ is the unique map with

$$\begin{array}{ccc}
 A_1 & \xrightarrow{\quad} & \bar{A}_1 \\
 x \downarrow & & \downarrow \bar{x} \\
 A_2 & \xrightarrow{\quad} & \bar{A}_2
 \end{array}$$

- 1.814 most reflective subcategories are full; fullness is necessary and sufficient for the reflection functor to be idempotent.

- 1.815 **CLOSURE OPERATION**: the reflection functor of a full reflective subcategory P' of a poset P — order-preserving, idempotent, inflationary. $x \leq y \implies \bar{x} \leq \bar{y}$, $\bar{\bar{x}} = \bar{x}$, $x \leq \bar{x}$. Conversely its values form a full reflective subcategory.

- 1.816 **COREFLECTIVE** $A' \subset A$: the inclusion has a right-adjoint.

- 1.817 G has a left-adjoint F iff every $(A, G(-))$ is representable, by FA ; Fx is then the unique map with

$$\begin{array}{ccc} (A_2, G(-)) & \xrightarrow{\simeq} & (FA_2, -) \\ (x, G(-)) \downarrow & & \downarrow (Fx, -) \\ (A_1, G(-)) & \xrightarrow{\simeq} & (FA_1, -) \end{array}$$

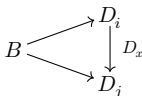
- 1.818 contravariant $F : A \rightarrow B$, $G : B \rightarrow A$ are **ADJOINT ON THE RIGHT** if $(B, FA) \simeq (A, GB)$ naturally, **ADJOINT ON THE LEFT** if $(FA, B) \simeq (GB, A)$. Between posets, adjoint on the right = Galois connection.

adjoint on the right iff F then $B \rightarrow B^\circ$ is left-adjoint of $B^\circ \rightarrow B$ then G ; adjoint on the left iff $A^\circ \rightarrow A$ then F is left-adjoint of G then $A \rightarrow A^\circ$. Covariant theorems thus yield contravariant ones.

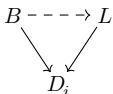
- 1.82 B^D : objects the functors $D \rightarrow B$, maps the natural transformations; D usually small. **DIAGONAL FUNCTOR** $\Delta : B \rightarrow B^D$: $B \mapsto$ the functor constant at B , $x \mapsto$ the transformation constant at x .

$D \neq \emptyset$: Δ is an embedding, B the subcategory of constant functors. $D = \emptyset$: B^D is the terminator among categories, and Δ has a right-adjoint iff B has a terminator, a left-adjoint iff B has a coterminator.

- 1.821 objects of B^D are diagrams modelled on D , whose objects i, j, k are the vertices. A map $\Delta(B) \rightarrow D$ is a *lower-bound*: a family $\{B \rightarrow D_i\}_{i \in |D|}$ compatible with every $x : i \rightarrow j$ of D .



D has a coreflection in Δ iff it has a greatest lower-bound $\{L \rightarrow D_i\}$: every lower-bound has a unique map over it, all i .



Discrete D : compatibility evaporates, greatest lower-bound = product diagram.

- 1.822 **LIMITS** $\lim D$: the values of a right-adjoint of $\Delta : B \rightarrow B^D$. **COLIMITS** $\text{colim } D$: the values of a left-adjoint — least upper bounds.

$\lim \emptyset$ is a terminator, $\text{colim } \emptyset$ a coterminator. For D finitely presented by $\cdot \rightrightarrows \cdot$ (two objects, two non-identity maps) the right-adjoint delivers equalizers, the left-adjoint coequalizers.

- 1.823 **COMPLETE**: every small diagram has a limit. **COCOMPLETE**: every small diagram has a colimit.

- 1.824 a family of monics $\{A_i \rightarrowtail A\}_I$ is a diagram modelled on I' (I discrete plus a terminator). Its limit is more than the intersection in $\mathcal{S}ub(A)$: any $B \rightarrow A$, monic or not, factors through $\lim A_i$ iff it factors through every A_i .

- 1.825 *complete iff equalizers and arbitrary products* dually *cocomplete iff coequalizers and arbitrary coproducts* and *(co-)cartesian iff every finite diagram has a (co-)limit*.

- 1.827 **CONTINUOUS**: preserves all small limits — from a complete source, iff it preserves equalizers and arbitrary products. **COCONTINUOUS**: preserves all small colimits, from a cocomplete source iff it preserves coequalizers and arbitrary coproducts. A contravariant functor is continuous if it carries colimits to limits.

- 1.828 *weak-* removes uniqueness. **WEAK-LIMIT** of D : a compatible $\{W \rightarrow D_i\}$ admitting, for every compatible $\{B \rightarrow D_i\}$, at least one

$$\begin{array}{ccc}
 B & \dashrightarrow & W \\
 & \searrow \quad \swarrow & \\
 & D_i &
 \end{array}$$

WEAKLY-COMPLETE: every small diagram has a weak-limit.

a functor preserving one weak-limit of a diagram preserves them all; a continuous functor from a complete category preserves all weak-limits.

- 1.829 avoid “weakly continuous”: a *functor that preserves weak-limits preserves limits* — preserving weak-limits already forces preservation of monic families.

- 1.82(10) **PRE-LIMIT** for $D : \mathbf{D} \rightarrow \mathbf{B}$: a set of lower-bounds cofinal in the class of all of them — compatible families $\{\{P_j \rightarrow D_i\}_{|D|}\}_J$ such that every lower-bound $\{B \rightarrow D_i\}$ admits $j \in J$ with

$$\begin{array}{ccc}
 B & \dashrightarrow & P_j \\
 & \searrow \quad \swarrow & \\
 & D_i &
 \end{array}$$

for all $i \in |D|$.

PRE-COMPLETE: every small diagram has a pre-limit. Preserving pre-limits $\implies$ preserving weak-limits $\implies$ [1.829] preserving limits.

- 1.83 **PRE-ADJOINT** for $A \in \mathbf{A}$, given $G : \mathbf{B} \rightarrow \mathbf{A}$: a set $\{A \rightarrow G(B_i)\}_I$ such that every $A \rightarrow G(B)$ admits $i \in I$, $x : B_i \rightarrow B$ with

$$\begin{array}{ccc}
 & A & \\
 \swarrow & & \searrow \\
 G(B_i) & \dashrightarrow & G(B) \\
 & G(x) &
 \end{array}$$

Locally small $\mathbf{A}$: a set $\{B_i\}_I$ in $\mathbf{B}$ through one of which every $A \rightarrow G(B)$ factors this way.

$\mathbf{B}$ full, G its inclusion: a pre-adjoint is a **PRE-REFLECTION** of A — a set $\{B_i\}$ through one of which every map from A to a $\mathbf{B}$ -object factors.

PRE-ADJOINT FUNCTOR: every $A \in \mathbf{A}$ has a pre-adjoint.

THE GENERAL ADJOINT FUNCTOR THEOREM *If $\mathbf{B}$ is locally small and complete then $G : \mathbf{B} \rightarrow \mathbf{A}$ has a left-adjoint iff it is continuous and pre-adjoint.*

- 1.831 **UNIFORMLY CONTINUOUS** $G : \mathbf{B} \rightarrow \mathbf{A}$: for every small $D : \mathbf{D} \rightarrow \mathbf{B}$, G carries the lower-bounds of D to a cofinal family of lower-bounds of $\mathbf{D} \xrightarrow{D} \mathbf{B} \xrightarrow{G} \mathbf{A}$ — every $\{A \rightarrow G(D_i)\}$ admits a lower-bound $\{B \rightarrow D_i\}$ and

$$\begin{array}{ccc} A & \dashrightarrow & G(B) \\ & \searrow & \swarrow \\ & G(D_i) & \end{array}$$

for all i .

such G preserves pre-limits, hence weak-limits, hence limits. $\mathbf{B}$ pre-complete: uniformly continuous = preserves pre-limits. $\mathbf{B}$ (weakly) complete: = preserves (weak) limits.

THE MORE GENERAL ADJOINT FUNCTOR THEOREM *If $\mathbf{B}$ is locally small and idempotents split in $\mathbf{B}$ then G has a left-adjoint iff it is uniformly continuous and pre-adjoint.*

- 1.832 **POINTWISE CONTINUOUS** $T : \mathbf{B} \rightarrow \mathcal{S}$: for every small D , every lower-bound $\{1 \rightarrow T(D_i)\}$ in $\mathcal{S}$ comes from a lower-bound $\{B \rightarrow D_i\}$ in $\mathbf{B}$ and a point

$$\begin{array}{ccc} 1 & \dashrightarrow & T(B) \\ & \searrow & \swarrow \\ & T(D_i) & \end{array}$$

G is uniformly continuous iff every $(A, G(-))$ is pointwise continuous.

- 1.833 smallest subfunctor $T' \subset T$ containing $\{x_i \in T(B_i)\}_I$: $y \in T'(B)$ iff $(Tf)(x_i) = y$ for some $i \in I$, $f : B_i \rightarrow B$. $T' = T$: T is generated by the x_i .
PETTY-FUNCTOR: generated by some set of elements. G is pre-adjoint iff every $(A, G(-))$ is petty.

- 1.834 **THE GENERAL REPRESENTABILITY THEOREM** *If $\mathbf{B}$ is locally small and idempotents split in $\mathbf{B}$ then $T : \mathbf{B} \rightarrow \mathcal{S}$ is representable iff it is a pointwise continuous petty-functor.*

1.835 *A locally small category in which idempotents split has a coterminator iff it has a pre-coterminator and all small diagrams have lower-bounds.*

1.836 $\hat{B} = \text{Split}(B)$ formally splits every idempotent; $B \rightarrow \hat{B}$ is full, uniformly continuous and pre-adjoint, with a left-adjoint only where idempotents already split. *A set-valued functor from a locally small category is a retract of a representable iff it is a pointwise continuous petty-functor.*

1.837 B complete: $\Delta : B \rightarrow B^D$ is continuous, and pre-adjoint iff all diagrams modelled on D have pre-colimits. Hence *a complete locally small category is cocomplete iff it is pre-cocomplete.*

1.838 **WELL-POWERED** B : $\text{Su}\mathcal{B}(B)$ is a set for every B . Distinct maps have distinct graphs, so this implies local smallness.

minimal object: no proper subobject (an atom is not minimal — it has exactly one). *Each object of a complete well-powered category has a unique minimal subobject.*

minimal objects form an initial class $\mathcal{M}$, pre-ordered ($f, g : M \rightarrow A$ have a non-proper equalizer, so $f = g$), every subset of which has a lower bound; an initial set exists iff $\mathcal{M}$ has one. Any map between minimal objects is a cover.

1.839 pre-adjoints come from a *cardinality function* K : objects to cardinals, with $K^{\#(\alpha)}$ a set of isomorphism types for every cardinal α . For continuous $G : B \rightarrow A$, B complete well-powered: B is *generated by* A through G if $A \rightarrow G(B)$ factors through no $G(B') \subset G(B)$ with $B' \subsetneq B$.

1.83(10) **COGENERATING SET** $\{C_i\}_I$: the $\{(-, C_i)\}_I$ are collectively an embedding — $f \neq g : A \rightarrow B$ gives $i \in I$, $h : B \rightarrow C_i$ with $fh \neq gh$. In a complete category: iff every object is embeddable in a product of the C_i .

THE SPECIAL ADJOINT FUNCTOR THEOREM *If B is complete, well-powered and has a cogenerating set, and A is locally small, then every continuous $G : B \rightarrow A$ has a left-adjoint.*

A complete well-powered category with a cogenerating set has a coterminator — the minimal subobject of $\Pi_I C_i$.

- 1.83(11) *dually: if $\mathbf{A}$ is cocomplete, well-co-powered and has a generating set, every cocontinuous functor from $\mathbf{A}$ to a locally small category has a right-adjoint, and every contravariant continuous functor to a locally small category has an adjoint on the right.*

1.84 **GIRAUD DEFINITION** of a Grothendieck topos (the first of four): locally small, cocomplete, effective regular, pullbacks preserve arbitrary unions, disjoint unions of objects exist, and there is a generating set. Such is a pre-topos and a logoi.

- 1.842 *If $\mathbf{E}$ is a Grothendieck topos then the graphing functor $\mathbf{E} \rightarrow \mathcal{Rel}(\mathbf{E})$ has a right-adjoint. Via 1.843–6 and the special adjoint functor theorem; chapter 2 proves it directly.*

- 1.843 *A Grothendieck topos is well-powered: a generating set $\{G_i\}_I$ is a basis, every subobject being an equalizer, so $B' \subset B$ is fixed by $\{(G_i, B') \subset (G_i, B)\}_I$. Hence $\mathcal{Rel}(\mathbf{E})$ is locally small.*

comonics coincide with covers, and covers out of A correspond to the equivalence relations on A : a Grothendieck topos is well-copowered.

- 1.844 *A Grothendieck topos is locally complete — the union of $\{A_i \rightarrowtail A\}$ is the image of $\Sigma A_i \rightarrow A$, Σ the disjoint union. Inverse images preserve arbitrary unions, so composition of relations distributes with arbitrary unions.*

- 1.845 *coproduct $\{u_i : A_i \rightarrow S\}_I$: $u_i u_i^\circ = 1$, $u_i u_j^\circ = 0$ for $i \neq j$, $\bigcup u_i^\circ u_i = 1_S$ — arbitrary coproducts are disjoint unions. A coproduct in $\mathbf{E}$ remains a coproduct in $\mathcal{Rel}(\mathbf{E})$: $R = \bigcup u_i^\circ R_i$ is the unique relation with $u_i R = R_i$, all i .*

- 1.846 *A coequalizer in $\mathbf{E}$ remains a coequalizer in $\mathcal{Rel}(\mathbf{E})$. A Grothendieck topos is $\mathbf{E}$ -standard: the smallest equivalence relation containing R is $\bigcup_{n=-\infty}^{\infty} R^n$, with $R^0 = 1$, $R^{-n} = (R^\circ)^n$.*

- 1.847 *Rel the ordering relation on the two-element ordered set is idempotent in $\mathcal{Rel}(\mathcal{S})$ but cannot be split. So $\mathcal{Rel}(\mathbf{E})$ is not cocomplete, though disjoint unions*

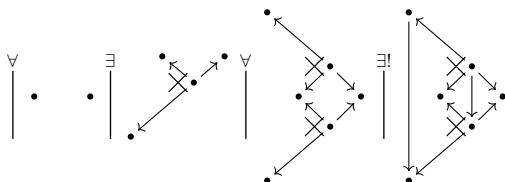
give it coproducts — and, reciprocation making it self-dual, products too.

- 1.85 **EXPONENTIAL $\mathbf{A}$** with binary products (elsewhere: cartesian closed): every $A \times -$ has a right-adjoint; equivalently every $(A \times -, B)$ is representable, by an object B^A .

an equivalence $\eta : (-, B^A) \rightarrow (A \times -, B)$ carries $1 \in (B^A, B^A)$ to an **EVALUATION MAP** $e : A \times B^A \rightarrow B$, and every $f : A \times X \rightarrow B$ has a unique $\lambda f : X \rightarrow B^A$ with

$$\begin{array}{ccc} & A \times X & \\ 1 \times \lambda f \swarrow & & \searrow f \\ A \times B^A & \xrightarrow{e} & B \end{array}$$

in the diagrammatic language — the cross marking a product diagram:



- 1.852 a poset is exponential iff it has binary intersections and every a, b have b^a with $x \leq b^a$ iff $a \wedge x \leq b$ — the Heyting arrow $a \rightarrow b$ of 1.72.

- 1.853 B^A is a bifunctor $\mathbf{A}^\circ \times \mathbf{A} \rightarrow \mathbf{A}$, with $f^A = \lambda \left(A \times B_1^A \xrightarrow{e} B_1 \xrightarrow{f} B_2 \right)$ and $B^g = \lambda \left(A_1 \times B^{A_2} \xrightarrow{g \times 1} A_2 \times B^{A_2} \xrightarrow{e} B \right)$:

$$\begin{array}{ccc} B_1^{A_2} & \xrightarrow{f^{A_2}} & B_2^{A_2} \\ B_1^g \downarrow & & \downarrow B_2^g \\ B_1^{A_1} & \xrightarrow{f^{A_1}} & B_2^{A_1} \end{array}$$

$(-)^A$ has the left-adjoint $A \times -$, so preserves limits. Commutativity of products makes $B^{(-)}$ adjoint on the right to itself, $(A_1, B^{A_2}) \simeq (A_2 \times A_1, B) \simeq (A_2, B^{A_1})$, hence continuous: colimits to limits.

bicartesian exponential: $1^A \simeq 1$, $(B_1 \times B_2)^A \simeq B_1^A \times B_2^A$, $B^0 \simeq 1$, $B^{A_1+A_2} \simeq B^{A_1} \times B^{A_2}$, $(B^{A_1})^{A_2} \simeq B^{A_1 \times A_2}$, $B^1 \simeq B$.

$A \times -$ has a right-adjoint, so preserves colimits: $A \times (B_1 + B_2) \simeq (A \times B_1) + (A \times B_2)$, $A \times 0 \simeq 0$ — the last equivalent to strictness of the cotermimator.

caution: 0^A is not 0, and $0^0 \simeq 1$. $(-)^A$ preserves monics, so 0^A is a subterminator: the largest U with $A \times U = 0$.

1.854 in any category with binary products, $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ is a left-adjoint of $\Delta : \mathbf{A} \rightarrow \mathbf{A}/B$: $(\Sigma f, C) \simeq (f, \Delta C)$, with $f : A \rightarrow B$ a slice object and ΔC the projection $B \times C \rightarrow B$.

a right-adjoint of Δ , where it exists, is written $\Pi : \mathbf{A}/B \rightarrow \mathbf{A}$: products of families in $\mathbf{A}$ indexed on B .

If $\Pi : \mathbf{A}/B \rightarrow \mathbf{A}$ exists then A^B exists for every A : $(B \times -, A) \simeq (\Sigma \Delta(-), A) \simeq (\Delta(-), \Delta A) \simeq (-, \Pi \Delta(A))$.

If $\mathbf{A}$ is cartesian and exponential then $\Delta : \mathbf{A} \rightarrow \mathbf{A}/B$ has a right-adjoint Π for every B — for $f : A \rightarrow B$ take the pullback

$$\begin{array}{ccc} \Pi(f) & \xrightarrow{\quad} & A^B \\ \downarrow & \lrcorner & \downarrow f^B \\ 1 & \xrightarrow{\quad} & B^B \\ & \lambda 1_B & \end{array}$$

$(C, \Pi(f))$, as a subset of $(B \times C, A)$, is then the set of g with

$$\begin{array}{ccc} B \times C & \xrightarrow{g} & A \\ & \searrow f & \swarrow \\ & B & \end{array}$$

which is exactly $(\Delta(C), f)$.

1.856 the slice lemma fails for exponential categories: $\mathbf{P}$ the posets, $\mathbf{3}$ the three-element chain, $\mathbf{P}/3$ not

exponential. $\mathbf{P}/A$ is exponential iff A contains no three-element chain.

- 1.857 A full coreflective subcategory of an exponential category closed under binary products is exponential: $(A \times C, B) \simeq (C, B^A) \simeq (C, G(B^A))$ for the coreflection G .

EXPONENTIAL IDEAL: a full $\mathbf{A}' \subset \mathbf{A}$ with $B^A \in \mathbf{A}'$ for all $A \in \mathbf{A}$, $B \in \mathbf{A}'$. **REPLETE**

SUBCATEGORY: closed under isomorphism type.

A full replete reflective subcategory of an exponential category is an exponential ideal iff reflections preserve products — from $(C, B^A) \simeq (\overline{C}, B^A)$, every $C \rightarrow B^A$ with $B \in \mathbf{A}'$ extends uniquely:

$$\begin{array}{ccc} C & \xrightarrow{\quad} & \overline{C} \\ & \searrow & \swarrow \text{dashed} \\ & B^A & \end{array}$$

At $C = B^A$ this forces $C \rightarrow \overline{C}$ to be an isomorphism.

- 1.858 **KURATOWSKI INTERIOR OPERATION** on a lattice: deflationary, idempotent, intersection-preserving — $x^\bullet \leq x$, $x^\bullet = (x^\bullet)^\bullet$, $(x \wedge y)^\bullet = x^\bullet \wedge y^\bullet$. Its values are the *open* elements; they form a Heyting algebra whenever the ambient lattice does.

LAWVERE-TIERNEY CLOSURE OPERATION (L-T): inflationary, idempotent, intersection-preserving — $x \leq \overline{x}$, $\overline{\overline{x}} = \overline{x}$, $\overline{\overline{x} \wedge \overline{y}} = \overline{x} \wedge \overline{y}$. Its values are the *closed* elements.

The closed elements of an L-T closure operation on a Heyting algebra form an exponential ideal — $a \rightarrow b$ is closed whenever b is; conversely a closure operation whose closed elements form an exponential ideal is L-T. Double negation is one.

- 1.859 **BASEABLE** B , in a category with binary products: B^A exists — i.e. $(A \times -, B)$ is representable — for every A . The baseable objects form a full subcategory $\mathbf{B}$, with $f^A : B_1^A \rightarrow B_2^A$ defined as if all objects were baseable.

The inclusion $\mathbf{B} \rightarrow \mathbf{A}$ preserves equalizers: the equalizer of f^A, g^A is B_0^A . $\mathbf{B}$ is itself always exponential, since $(B^A)^{A'} \simeq B^{A \times A'}$.

1.9 Topoi

- 1.9 Any category with pullbacks composes a map with a relation: for $f : A \rightarrow B$ and a table $\langle R; b, c \rangle$ on B, C , pull b back along f .

$$\begin{array}{ccccc} R' & \longrightarrow & R & \xrightarrow{c} & C \\ \downarrow \lrcorner & & \downarrow b & & \\ A & \xrightarrow{f} & B & & \end{array}$$

$R' \rightarrow A$, $R' \rightarrow C$ is still a monic pair; that table is fR . No regularity needed, and $g(fR) = (gf)R$.

U targeted at C is a **UNIVERSAL RELATION** if every relation R targeted at C is uniquely fU ; that f is written ΛR . A **POWER-OBJECT** of C , written $[C]$, is the source of a universal relation, itself written $\exists \subset [C] \times C$.

$$\begin{array}{ccccc} R & \longrightarrow & \exists & \longrightarrow & C \\ \downarrow \lrcorner & & \downarrow & & \\ A & \xrightarrow{\Lambda R} & [C] & & \end{array}$$

A **TOPOS** is a cartesian category in which each object has a power-object. A *topos* is a *positive effective transitive logos*, hence a pre-topos.

- 1.911 $\mathcal{R}el(-, B)$ is contravariant set-valued on any category with pullbacks, $\mathcal{R}el(f, B)$ sending R to fR . $[B]$ exists iff $\mathcal{R}el(-, B) \simeq (-, [B])$. A regular $\mathbf{A}$ is a topos iff the graphing functor $\mathbf{A} \rightarrow \mathcal{R}el(\mathbf{A})$ [1.564] has a right adjoint. In $\mathcal{S}$, $[B]$ is the set of subsets of B .
- 1.912 $\mathcal{R}el(-, 1) \simeq \mathcal{S}ub(-)$, so $[1]$ — traditionally Ω — is a **SUBOBJECT CLASSIFIER**. A universal subobject is a monic $\Omega' \rightarrowtail \Omega$ of which every monic $A' \rightarrowtail A$ is a unique pullback; χ_A is the **CHARACTERISTIC MAP** of $A' \subset A$, equivalently the unique χ with $\chi^\#(\Omega') = A'$.

$$\begin{array}{ccc} A' & \rightarrowtail & A \\ \downarrow \lrcorner & & \downarrow \chi_A \\ \Omega' & \rightarrowtail & \Omega \end{array}$$

Uniqueness of χ_A forces uniqueness of $A' \rightarrow \Omega'$.

Taking $A' \rightarrowtail A$ to be 1_A gives a unique pullback for every A , so each object has exactly one map to Ω' : the universal subobject is a point, henceforth written $1 \xrightarrow{t} \Omega$.

1.913 *In any cartesian category with a subobject classifier every subobject is an equalizer — $A' \rightarrowtail A$ equalizes χ_A and $p_A t$. Consequently covers coincide with epics.*

1.914 An n -ary $g : \Omega^n \rightarrow \Omega$ induces $\hat{g}$ on each $\mathcal{Su}\mathcal{B}(A)$: $\hat{g}(A_1, \dots, A_n)$ is the subobject with characteristic map $\langle \chi_{A_1}, \dots, \chi_{A_n} \rangle g$. Every operation on $\mathcal{Su}\mathcal{B}(-)$ so arises, and each commutes with inverse images: $\hat{g}(f^\# A_1, \dots, f^\# A_n) = f^\# \hat{g}(A_1, \dots, A_n)$.

$g = \chi$ of $\langle t, t \rangle : 1 \rightarrow \Omega \times \Omega$ gives $\hat{g}(A_1, A_2) = A_1 \cap A_2$: Ω is a semi-lattice with unit $1 \rightarrow \Omega$. $g = \chi$ of $\langle 1, 1 \rangle : \Omega \rightarrow \Omega \times \Omega$ makes $\hat{g}(A_1, A_2)$ the equalizer of χ_{A_1}, χ_{A_2} , so $A' \subset \hat{g}(A_1, A_2)$ iff $A_1 \cap A' = A_2 \cap A'$ — the Heyting $\leftrightarrow$, with $x \rightarrow y = x \leftrightarrow (x \wedge y)$. Each $\mathcal{Su}\mathcal{B}(A)$ is a Heyting semi-lattice.

1.919 *Every monic endomorphism of Ω is an involution.* Absence of non-automorphic monic endomorphisms is a finiteness condition; Ω need not satisfy the dual condition.

1.91(10) *A nonempty category with binary products, equalizers and power-objects has a terminator, hence is a topos — the equalizer of $1_{[B]}$ and the constant $\Lambda M_{[B], B}$, where $M_{A, B}$ is the relation tabulated by $A \times B$.*

1.92 $\Lambda 1 : B \rightarrowtail [B]$ is the **SINGLETON MAP** in $\mathcal{S}$ it sends b to $\{b\}$. For any $f : A \rightarrow B$, $(f \Lambda 1) \ni = f$ and $f \Lambda 1$ is a map, hence $f \Lambda 1 = \Lambda f$; consequently $\Lambda 1$ is monic.

A topos is exponential. Power-objects are baseable, $(A \times -, [B]) \simeq \mathcal{Su}\mathcal{B}(- \times A \times B) \simeq (-, [A \times B])$, so $[B]^A$ is constructible as $[A \times B]$; and B is the equalizer of the characteristic map χ of $\Lambda 1$ and of pt , hence baseable [1.859].

$$B \xrightarrow{\Lambda 1} [B] \begin{array}{c} \xrightarrow{\chi} \\ \xleftarrow{pt} \end{array} \Omega$$

1.921 $\mathcal{R}el(-, B) \simeq \mathcal{S}ub(- \times B) \simeq (- \times B, \Omega) \simeq (-, \Omega^B)$: $[B]$ is definable as Ω^B . Lawvere's original axioms — bicartesian, exponential, with partial map classifiers [1.934] — follow from power-objects alone.

1.922 $[-]$ is the covariant power-object functor, right adjoint of the graphing functor; Ω^- is contravariant. For $g : B_1 \rightarrow B_2$, Ω^g is the unique map inducing $R \mapsto Rg^\circ$ from $\mathcal{R}el(-, B_2)$ to $\mathcal{R}el(-, B_1)$; so $\Omega^g = [g^\circ]$.

1.923 B^A is cut out of $[A \times B]$ by a pullback.

$$\begin{array}{ccc} B^A & \xrightarrow{\quad} & [A \times B] \\ \downarrow & \lrcorner & \downarrow f \\ 1 & \xrightarrow{\quad} & [A] \end{array} \quad \text{“} \ulcorner_A \text{”}$$

In $\mathcal{S}$, f sends $R \subset A \times B$ to $\{a \mid \exists! b, \langle a, b \rangle \in R\}$, and $1 \rightarrow [A]$ names the entire subobject.

1.926 In a topos exponential structure restricts to the Heyting algebra structure on $\mathcal{S}ub(1)$.

1.93 Slice lemma for topoi: for a topos $\mathbf{E}$ and an object B , $\mathbf{E}/B$ is a topos and $\Delta : \mathbf{E} \rightarrow \mathbf{E}/B$ is a representation of topoi. $\Delta[A]$ is the power-object of $\Delta(A)$; and for $A' \subset A$, $[A']$ is the equalizer of $1_{[A]}$ and the idempotent $e = \Lambda(\exists \cap ([A] \times A'))$.

1.931 In any cartesian category $f : A \rightarrow B$ gives $f^\# : \mathbf{A}/B \rightarrow \mathbf{A}/A$ by pulling back, with left adjoint Σ_f by composing.

$$\begin{array}{ccc} f^\# C & \xrightarrow{\quad} & C \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{\quad} & B \end{array} \quad \text{“} \ulcorner_f \text{”}$$

THE FUNDAMENTAL LEMMA OF TOPOI For $f : A \rightarrow B$ in a topos $\mathbf{E}$, $f^\# : \mathbf{E}/B \rightarrow \mathbf{E}/A$ has a right adjoint $\Pi_f : \mathbf{E}/A \rightarrow \mathbf{E}/B$.

1.932 $f^\#$ restricted to $\mathcal{S}ub(B)$ is inverse image, so Π_f restricted to $\mathcal{S}ub(A)$ is the double-sharp operation [1.7]. The double-sharp axiom holds for topoi.

1.933 Pullbacks preserve covering families: for $f : A \rightarrow B$ and a cover $\{C_i \rightarrow B\}_I$, the $\{D_i \rightarrow A\}$ below are a cover.

$$\begin{array}{ccc}
D_i & \longrightarrow & C_i \\
\downarrow & \lrcorner & \downarrow \\
A & \longrightarrow & B
\end{array}
\quad f$$

Consequently a topos is pre-regular.

- 1.934 $\mathcal{P}ar(\mathbf{C})$ has the objects of $\mathbf{C}$, and as morphisms $A \rightarrow B$ the pairs $\langle A', f \rangle$ with $A' \subset A$, $f: A' \rightarrow B$; composition of $\langle A', f \rangle$ and $\langle B', g \rangle$ is $\langle f^\#(B'), fg \rangle$.

For a topos $\mathbf{E}$ the inclusion $\mathbf{E} \subset \mathcal{P}ar(\mathbf{E})$ has a right adjoint: $\mathcal{P}ar(-, A) \simeq (-, \tilde{A})$.

$$\begin{array}{ccc}
B' & \twoheadrightarrow & B \\
f \downarrow & \lrcorner & \downarrow \\
A & \twoheadrightarrow & \tilde{A}
\end{array}$$

$\tilde{A} = \Pi_t(A)$; $\tilde{1} = \Omega$, and in a boolean positive logoss $\tilde{A} = A + 1$.

- 1.935 Topoi are pre-regular and satisfy the slice condition [1.541]. Every topos may be faithfully represented in a capital topos.

- 1.94 For $F \subset [A]$, $\Gamma(F)$ is a subset of $\mathcal{S}ub(A)$: the **SUBOBJECTS NAMED BY F** . The **INTERNALLY DEFINED INTERSECTION** $\cap F$:

$$\begin{array}{ccc}
\cap F & \twoheadrightarrow & A \\
\downarrow & \lrcorner & \downarrow \\
1 & \longrightarrow & \Omega^F
\end{array}$$

$A \rightarrow \Omega^F$ is the adjoint of $F \twoheadrightarrow [A] \simeq \Omega^A$, and $1 \rightarrow \Omega^F$ the adjoint of $F \rightarrow 1 \rightarrow \Omega$.

- 1.941 $\cap F$ is preserved by representations of topoi, and is the largest subobject of A that stays a lower bound of the subobjects named by $T(F)$ under every representation T . “Internally defined” is coextensive with being preserved by representations.

- 1.942 The adjoint $1 \rightarrow \Omega^A$ of $\chi_{A'}$ is the **NAME OF A'** , written $\ulcorner A' \urcorner$; A' is named by F iff $\ulcorner A' \urcorner \subset F$, and the $1 \rightarrow \Omega^F$ of $\cap F$ is the name of the entire subobject. Ω^f is the internally defined inverse image: $\ulcorner B' \urcorner \Omega^f = \ulcorner f^\# B' \urcorner$.

For $A' \subset A$ named by F , the composite $A \rightarrow \Omega^F$ followed by $\Omega^{\ulcorner A' \urcorner}$ is $\chi_{A'}$.

-
- 1.943 $\cap F$ is contained in every subobject named by F . If F is well-pointed, $\cap F$ is their greatest lower bound, and $B \rightarrow A$ factors through $\cap F$ iff it factors through each subobject named by F .
-
- 1.944 A topos has a strict coterminal — $\cap \Omega$ in $\mathcal{Su}\mathcal{B}(1)$, minimal and stably so under every $p^\#$.
-
- 1.945 A topos is regular. Image of $f : A \rightarrow B$: let $F \subset [B]$ name the B' with $f^\#(B') = A$; then $\cap F = \text{Im } f$.
-
- 1.946 A topos is a logos. Binary unions: $F_i \subset [A]$ names the A' containing A_i ; $F = F_1 \cap F_2$ names those containing both, and $\cap F = A_1 \cup A_2$.
-
- 1.947 A topos is a transitive logos. For reflexive R on B : F_1 , the equalizer of 1 and $[R \times 1]$ in $[B \times B]$, names the S with $RS \subset S$; F_2 names the reflexive ones; $\cap (F_1 \cap F_2) = R^*$.
-
- 1.949 The **INTERNALLY DEFINED UNION** $\cup F$ is the direct image $\mu(\exists \cap (F \times A))$ along $\mu : [A] \times A \rightarrow A$. It is an upper bound of every subobject named by F , and the least one when F is well-pointed.
-
- $A' \subset A$ is a permanent lower (upper) bound of F if $T(A')$ stays one under every representation T . $\cap F$ is the greatest permanent lower bound, $\cup F$ the least permanent upper bound.
-
- 1.94(10) $\Lambda 1 : A \rightarrow [A]$ names the subobjects which are points; $\cup (\Lambda 1)$ is always entire, and is their least upper bound only when A is well-pointed.
-
- A subobject of A is its **WELL-POINTED PART** $A \cdot$ if it is the least upper bound of the points of A , equivalently the greatest well-pointed subobject; no internal construction of $A \cdot$ is possible. A **SOLVABLE TOPOS** is one in which every object has a well-pointed part; there $\mathcal{Su}\mathcal{B}(A)$ has bounds for every elementarily definable family, but the slice lemma fails. Capital topoi are solvable, so every topos is faithfully represented in a solvable one.
-
- 1.95 A topos is a pre-topos.
-
- 1.951 A topos is effective. For an equivalence relation E on A , factor ΛE as $A \xrightarrow{h} B \rightarrowtail [A]$; then $h^\circ h = 1$ and $hh^\circ = E$.
-
- 1.952 A topos is positive.
-

$$\begin{array}{ccc}
0 & \longrightarrow & 1 \\
\downarrow & \lrcorner & \downarrow \\
A & \xrightarrow{\Lambda 1} & [A]
\end{array}$$

since ΛR factors through $\Lambda 1$ iff R is a map, and through $\lrcorner 0 \rceil$ iff $R = 0$. So $A + 1$ exists; generally $\langle \Lambda 1, \lrcorner 0 \rceil$ and $\langle \lrcorner 0 \rceil, \Lambda 1 \rangle$ are disjoint monics into $[A] \times [B] \simeq [A + B]$, tabulating $A + B$.

- 1.954–5 *A topos has coequalizers*: for $f, g : A \rightarrow B$ put $R = f^\circ g$ and $S = (R \cup R^\circ)^*$; effectiveness makes S the level of some $B \rightarrow C$, the coequalizer. *A topos is bicartesian* (coequalizers, coterminal, binary coproducts).

- 1.961 *E* is an **INJECTIVE** if $(-, E)$ carries monics to epics.

$$\begin{array}{ccc}
A & \twoheadrightarrow & B \\
\downarrow & & \swarrow \\
E & \hookrightarrow &
\end{array}$$

In a pre-topos, pushouts of monics are monic [1.651], so E is injective iff every monic $E \rightarrowtail A$ has a right inverse.

E in an exponential category is **INTERNALLY INJECTIVE** if E^- carries monics to epics. *In a topos Ω is internally injective*: for monic f , $\Omega^f = [f^\circ]$ has left inverse $[f]$, since $\Omega^f[f] = [f^\circ f] = 1$. *In a topos Ω is (externally) injective*.

- 1.962 *If E is injective in an exponential category then so is E^A* , since $(-, E^A) \simeq (- \times A, E)$ and $- \times A$ preserves monics; likewise internally. Hence Ω^A is injective, and $\Lambda 1$ embeds every object in an injective.

- 1.963 *$\tilde{A}$ [1.934] is injective* — a partial map $B_1 \rightarrow A$ is a partial map $B_2 \rightarrow A$ whenever $B_1 \rightarrowtail B_2$.

- 1.964 A category is **VALUE-BASED** if its values form a basis; capital topoi are. *In a value-based topos Ω is a cogenerator*.

- 1.965 *C* in an exponential category **INTERNALLY COGENERATES** if C^- is a contravariant embedding;

-
- a cogenerator does. *In a topos Ω internally cogenerates.*
-
- 1.966 A **PROGENITOR** is an object whose subobjects form a generating set. A topos is value-based iff 1 is a progenitor; every Grothendieck topos has one. *If G is a progenitor then Ω^G is a cogenerator.*
-
- 1.967 *Equivalent on a locally small topos: (a) arbitrary powers exist; (b) arbitrary copowers exist; (c) arbitrary copowers of 1 exist. Each implies local completeness.* (a) says contravariant representables have adjoints, (b) that covariant ones do, (c) that Γ does.
-
- 1.968 *A locally small topos is complete iff it is cocomplete.*
-
- 1.969 The **LAWVERE DEFINITION** of a Grothendieck topos: a cocomplete topos with a generating set. The **TIERNEY DEFINITION**: a topos with a progenitor and arbitrary copowers of 1 — equivalently, with a geometric morphism to $\mathcal{S}$.
-
- 1.96(11) *Slice lemma for Grothendieck topoi: $\mathbf{E}/B$ is a Grothendieck topos, and $\Delta : \mathbf{E} \rightarrow \mathbf{E}/B$ preserves progenitors.*
-
- 1.971 A is small if it is a subquotient of a $\mathbf{G}$ -object: $A \leftarrow G' \twoheadrightarrow G$ with the left arrow a cover. *In a boolean Grothendieck topos every object is a coproduct of small objects.*
-
- 1.972 *In a boolean logos with 1 projective, 1 is a progenitor. Equivalent on a Grothendieck topos: (1) boolean and 1 projective; (2) boolean and value-based; (3) boolean with a choice progenitor; (4) every object a coproduct of sub-terminators; (5) AC.*
-
- 1.973 A topos is **IAC** (internal axiom of choice) if $(-)^A$ [1.853] preserves epics for every A .
-
- 1.974 *A topos is AC iff it is IAC and 1 is projective.* Every functor from an AC topos preserves epics, since it preserves left-invertibility.
-
- 1.976 *A small topos may be faithfully represented in an AC topos iff it is IAC* — take $C_A \subset A^{[A]^+}$ naming the choice maps contained in the universal relation.
-

- 1.977 Any universal sentence in the predicates of topoi which follows from AC follows from IAC. In particular an IAC topos is boolean.
- 1.979 For any topos $\mathcal{A}$ there is a boolean topos $\mathcal{B}$ and a faithful bicartesian representation $\mathcal{A} \rightarrow \mathcal{B}$ [2.542].
- 1.98 A **NATURAL NUMBERS OBJECT** (NNO) is $1 \xrightarrow{0} N \xrightarrow{s} N$ such that every $1 \xrightarrow{a} A \xrightarrow{t} A$ admits a unique $N \rightarrow A$:

$$\begin{array}{ccccc}
 1 & \xrightarrow{0} & N & \xrightarrow{s} & N \\
 & \searrow a & \downarrow & & \downarrow \\
 & & A & \xrightarrow{t} & A
 \end{array}$$

Unique up to isomorphism. In $\mathcal{S}$: zero and successor. The topos of finite sets has no NNO.

- 1.981 For a NNO and any $A \xrightarrow{x} B \xrightarrow{t} B$ there is a unique $A \times N \rightarrow B$:

$$\begin{array}{ccccc}
 A & \xrightarrow{\langle 1_A, 0 \rangle} & A \times N & \xrightarrow{1_A \times s} & A \times N \\
 & \searrow x & \downarrow & & \downarrow \\
 & & B & \xrightarrow{t} & B
 \end{array}$$

— transpose x, t into $1 \rightarrow B^A \rightarrow B^A$.

- 1.983 *Recursion*: for a NNO and any $g : A \rightarrow B$, $h : A \times N \times B \rightarrow B$ there is a unique $f : A \times N \rightarrow B$ with $\langle 1_A, 0 \rangle f = g$ and $(1_A \times s)f = \langle p_1, p_2, f \rangle h$.
- 1.984 Addition, multiplication and exponentiation on a NNO by the usual recursion equations: $\langle 1_N, 0 \rangle a = 1_N$, $(1_N \times s)a = as$; $\langle 1_N, 0 \rangle m = 0$, $(1_N \times s)m = \langle p_1, m \rangle a$; $\langle 1_N, 0 \rangle e = 0s$, $(1_N \times s)e = \langle p_1, e \rangle m$.
- 1.985 For a NNO, $1 \xrightarrow{0} N \xleftarrow{s} N$ is a coproduct and $N \rightrightarrows N \rightarrow 1$ (by 1_N and s) is a coequalizer.

$$\begin{array}{ccc}
 1 & & N \\
 & \searrow 0 & \swarrow s \\
 & & N
 \end{array}$$

$$\begin{array}{ccc}
 & 1_N & \\
 N & \rightrightarrows & N \longrightarrow 1 \\
 & s &
 \end{array}$$

- 1.986 Given $1 \xrightarrow{a} A$ and $A \xrightarrow{f} A$ there is $A' \subset A$ carrying a', f' with

$$\begin{array}{ccccc}
 1 & \xrightarrow{a'} & A' & \xrightarrow{f'} & A' \\
 & \searrow a & \downarrow & & \downarrow \\
 & & A & \xrightarrow{f} & A
 \end{array}$$

and with $\left(\begin{smallmatrix} a' \\ f' \end{smallmatrix}\right) : 1 + A' \rightarrow A$ a cover — in $\mathcal{Rel}(\mathbf{A})$ take $A'' = af^*$, $f'' = A''f$.

- 1.987 $1 \xrightarrow{a} A$, $A \xrightarrow{t} A$ has the **PEANO PROPERTY** iff every subobject $B \subset A$ that allows a and allows $t \upharpoonright B$, the composite $B \subset A \xrightarrow{t} A$, is entire. The A' of 1.986 is the least subobject allowing both a and $A' \subset A \xrightarrow{f} A$, and it has the Peano property.

- 1.988 In a boolean topos, if $1 \xrightarrow{a} A \xleftarrow{t} A$ is a coproduct and $A \rightrightarrows A \rightarrow 1$ (by 1_A and t) a coequalizer, then $1 \xrightarrow{a} A \rightarrow A$ has the Peano property. (“Boolean” is removable by [2.542].)

- 1.989 Under the hypotheses of 1.988, if $\left(\begin{smallmatrix} c \\ u \end{smallmatrix}\right) : 1 + C \rightarrow C$ is a cover and $cf = a$, $uf = ft$, then $f : C \rightarrow A$ is monic.

- 1.98(10) Converse of 1.985: in any topos, if $1 \xrightarrow{a} A \xleftarrow{t} A$ is a coproduct and $A \rightrightarrows A \rightarrow 1$ a coequalizer, then $1 \xrightarrow{a} A \xrightarrow{t} A$ is a NNO. Uniqueness from the Peano property applied to the equalizer of two candidates; existence from 1.986 applied to $\langle a, b \rangle$ and $t \times v$.

- 1.98(11) A bicartesian functor $T : \mathbf{A} \rightarrow \mathbf{A}'$ of topoi carries a NNO to a NNO — by the bicartesian characterization [1.985, 1.98(10)].

- 1.98(12) An **A-ACTION** is an object B with $1 \xrightarrow{b} B$ and $A \times B \xrightarrow{v} B$. A **FREE A-ACTION** is an A -action $A \times 1 \xrightarrow{1_A \times e} A \times A^* \xrightarrow{s} A^*$ universal among them:

$$\begin{array}{ccccc}
 A \times 1 & \xrightarrow{1_A \times e} & A \times A^* & \xrightarrow{s} & A^* \\
 & \searrow 1_A \times b & \downarrow & & \downarrow f \\
 & & A \times B & \xrightarrow{v} & B
 \end{array}$$

Unique up to isomorphism; a NNO is a free 1-action. In $\mathcal{S}$, A^* is the set of finite lists, e the empty list, s prefixing.

-
- 1.98(13) $A \times 1 \rightarrow A \times A^* \rightarrow A^*$ is a free A -action iff $\binom{e}{s} : 1 + A \times A^* \rightarrow A^*$ is iso and $A \times A^* \rightrightarrows A^* \rightarrow 1$ is a coequalizer.
-
- 1.98(14) In a topos with a NNO every object A has a free A -action — take A^* the least subobject of $(1 + A)^N$ allowing e and $A \times A^* \subset A \times (1 + A)^N \rightarrow (1 + A)^N$.

1.(10) Sconing

- 1.(10) a category with a terminator is **EXACTING** if $\Gamma = (1, -)$ is exact, i.e. preserves whatever finite colimits exist (it always preserves whatever small limits exist). An exacting pre-logos is focal, the pasting lemma describing unions as pushouts; a Grothendieck topos is exacting iff 1 is a connected projective.

- 1.(10)1 *Every category $\mathbf{A}$ with a terminator is a slice of an exacting category $\hat{\mathbf{A}}$ — and regular / pre-logos / pre-topos / logos / cartesian closed / topos / Grothendieck topos / having a natural numbers object all transfer to $\hat{\mathbf{A}}$.*

the **SCONE** $\hat{\mathbf{A}}$: objects $\langle S, A, f \rangle$ with S a set and $f : S \rightarrow \Gamma(A)$, morphisms $\langle g, x \rangle$ over the square

$$\begin{array}{ccc} S & \xrightarrow{g} & S' \\ f \downarrow & & \downarrow f' \\ \Gamma(A) & \xrightarrow{\Gamma(x)} & \Gamma(A') \end{array}$$

written $\langle S, A \rangle$, the third coordinate being clear.

- 1.(10)11 $\langle \Gamma(1), 1 \rangle$ is the terminator; $M = \langle \emptyset, 1 \rangle$ is a proper subobject containing every proper subterminator, so 1 is no union of proper subobjects — it is coprime. $\hat{\mathbf{A}}/M \equiv \mathbf{A}$.

- 1.(10)12–13 the scone is exacting, $\Gamma : \hat{\mathbf{A}} \rightarrow \mathcal{S}$ preserving whatever small colimits exist. $\hat{\mathbf{A}} \rightarrow \hat{\mathbf{A}}/M \rightarrow \mathbf{A}$ has both adjoints — $A \mapsto \langle \emptyset, A \rangle$ left, $A \mapsto \langle \Gamma(A), A \rangle$ right — hence preserves all limits and colimits that exist.

- 1.(10)14 structure lifts coordinatewise:

$$\langle f, x \rangle \#\# \langle S, A \rangle = \langle f \#\# S, x \#\# A \rangle, \quad \langle T, B \rangle^{\langle S, A \rangle} = \langle \hat{\mathbf{A}}(\langle S, A \rangle, \langle T, B \rangle)$$

$$[\langle S, A \rangle] = \langle \mathcal{S}u\ell_{\hat{\mathbf{A}}} \langle S, A \rangle, [A] \rangle, \quad N_{\hat{\mathbf{A}}} = \langle N_{\mathcal{S}}, N_{\mathbf{A}} \rangle,$$

and $\mathcal{C}$ generating $\mathbf{A}$ makes $\{\langle 1, C \rangle : C \in \mathcal{C}\}$ generate $\hat{\mathbf{A}}$.

- 1.(10)3 for free $\mathbf{A}$ the composite $\mathbf{A} \rightarrow \hat{\mathbf{A}} \rightarrow \hat{\mathbf{A}}/M \rightarrow \mathbf{A}$ is the identity, so $\mathbf{A}$ is a **RETRACT** of $\hat{\mathbf{A}}$.

- 1.(10)31–2 *A retract of an exacting category is exacting, hence various free categories are exacting.*
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- 1.(10)4 *A in a cocomplete abelian category is a **SMALL PROJECTIVE** if $(A, -)$ preserves all colimits, equivalently all sums.*
-
- 1.(10)41 *For E a Grothendieck topos and A a connected projective, $(A, -) : E \rightarrow S$ preserves all small colimits.*